

POST-GROKING COLLAPSE AT THE REPRESENTATION–READOUT INTERFACE IN MUON-TRAINED TRANSFORMERS

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ABSTRACT

Post-grokking collapse can destroy accuracy while preserving decodable task information. We study modular-arithmetic transformers trained with Muon on hidden weights and AdamW on embeddings and the output head. At five captured unnormalized transitions, adjacent swaps localize the acute failure to the head update. Its mean-dependent logit term reproduces each failing test accuracy within 0.33 percentage points. Training-only linear decoders recover 98.20–100% at collapsed checkpoints. Corrected cross-entropy prevents joint train/test collapse through step 100,000 in five matched unnormalized branches; four prospective RMS-normalized runs still fail through embedding updates. The captured unnormalized updates locally descend accurate loss and overshoot at full scale. Finally, task-aligned Fourier filtering averages logits over inputs sharing the answer, allowing a memorizer to achieve perfect filtered accuracy.

1 INTRODUCTION

Generalization acquired during grokking can later collapse. In five captured failures of unnormalized modular-arithmetic transformers, swaps across a single optimizer update localize the acute damage to the readout: the preceding head remains accurate on the updated features, while the new head lowers accuracy on the preceding features. Fresh linear decoders fitted on training examples recover 98.20–100% held-out accuracy at selected failed checkpoints. The mean-dependent component of the actual readout displacement nearly reproduces each failure (Figure 1; §4.4).

Liu et al. (2026) already connect inaccurate cross-entropy derivatives, feature/classifier-mean growth, amplified Adam updates, and numerical corrections. Chou et al. (2026) use a representation–readout decomposition and linear probes to diagnose readout miscalibration during grokking. We add a joint account of individual post-grokking failures: adjacent-update localization, an intervention on the mean-dependent part of the measured readout displacement, and a test of the task information retained at failed states. These interventions identify the damaging update; the upstream contributions of feature growth and optimizer history remain coupled.

We use Muon (Jordan et al., 2024) with the routing used in large-scale training (Liu et al., 2025): Muon trains the hidden matrices, while AdamW trains the token and position embeddings and the readout. Our speed and stability comparisons concern the stated configurations, sweeps, and horizons. Matched corrected-cross-entropy continuations remain free of joint train/test failure through step 100,000 in all five unnormalized seeds; a tenfold post-grokking reduction in the hidden learning rate leaves failures in all five. With gain-free RMS, accurate-CE failures replicate in four prospective seeds. Swaps localize each captured transition to embeddings, while failed-state decoder accuracy varies substantially, establishing an architectural boundary for this account.

Fourier measurements support this analysis with a separate qualification. Retaining a full task-aligned family averages representations over inputs with the same answer, supplying enough task structure for a lookup-table memorizer to achieve perfect filtered accuracy. Raw-feature decoders and adjacent swaps assess surviving information independently of this projection. Spectral similarity during a captured collapse is discussed alongside the projection analyses in §6.

Contributions.

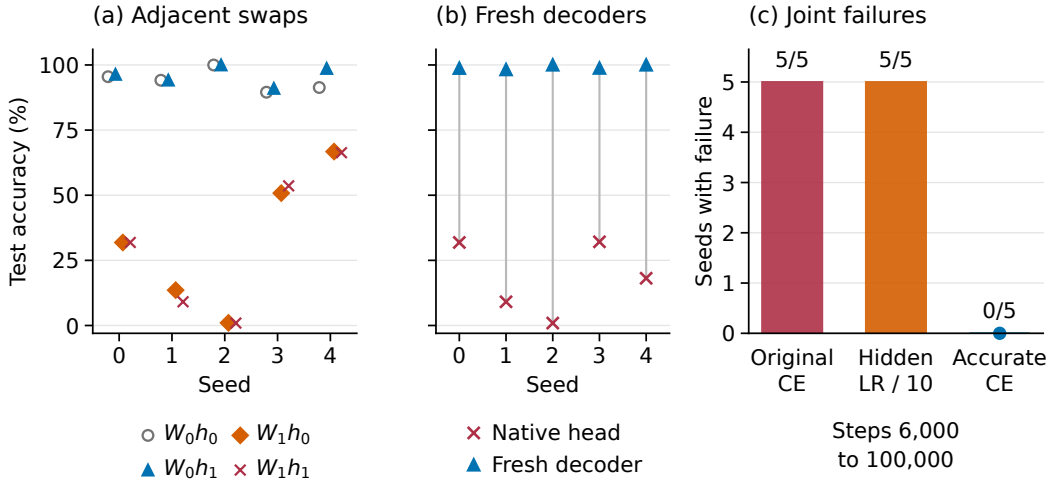


Figure 1: Follow-ups on five selected unnormalized trajectories; panels reuse seeds. **(a)** Adjacent swaps: 0/1 denote states before/after the captured update. The old head retains accuracy on new features; the new head lowers accuracy on old features. **(b)** Training-only decoders recover 98.20–100% at collapsed snapshots; seeds 3 and 4 use later snapshots than (a). **(c)** Joint train/test failures below 90%: original CE 5/5, hidden LR/10 5/5, accurate CE 0/5. Continuations share step-6,000 sources and end at 100,000. Binary incidence compares mixed original monitoring (failures captured before 30,000) with complete joint-event detection in both continuations (Appendix J).

- **Localized failures with decodable information.** At five captured unnormalized transitions, adjacent swaps isolate the readout update. Its mean-dependent displacement nearly reproduces the loss of accuracy, while fresh training-only decoders recover high held-out accuracy (§4).
- **Stabilization and its scope.** Matched continuations compare numerical correction and hidden-learning-rate reduction through 100,000 steps. Four prospective RMS runs establish limits to numerical prevention, readout localization, and retained decodability (§4.4).
- **Interpretation of Fourier interventions.** An orbit-average identity and memorizer controls explain the task structure supplied by filtering and calibrate its use as evidence for a learned algorithm (§5).

2 SETUP

Task. We train on $(a+b) \bmod p$, and on $(a-b) \bmod p$ where stated, using token sequences $[a, b, =]$ and p output classes. Defaults are $p = 113$ and a fixed random 30% training split of the p^2 operand grid. Training is full batch with cross-entropy. Original sweeps evaluate every 100 steps; diagnostic branches use finer grids where specified. The arithmetic follow-ups add the monitoring in §4.4.

Architecture. The model is a decoder-only transformer, $d_{\text{model}} = 128$, four heads, MLP width 512, ReLU, sequence length 3, every projection bias-free, and no normalization layers. An invertible change of residual basis can be absorbed by the matrices that write to and read from it. This is a symmetry of the parameterization; optimizer updates need not preserve it. The gain-free RMS control restricts this symmetry to $O(d)$. Depth is varied by appending blocks under nested initialization.

Routing and definitions. Parameters split into the three groups of Table 1; Muon receives the hidden group. Original grokking speeds use the first of five consecutive evaluations at or above 95% test accuracy. *Strictly stable* means no subsequent evaluation below 95%; we also report sub-threshold counts and minima. The depth-4 freeze and fresh follow-ups verify shared source states. Historical branch provenance is reported separately (Appendix A.1). CPU seed replications have identical logged metrics before the freeze (Appendix A). Follow-up confirmation uses six checks (Appendix J.1).

Table 1: Parameter groups and routing. The AdamW baseline places all parameters under one AdamW instance. Selected settings: Muon hidden $\text{lr} = 0.03$, $\text{wd} = 0.1$, $\beta = 0.95$, five Newton–Schulz iterations; auxiliary $\text{lr} = 10^{-3}$, $\text{wd} = 1.0$; readout $\text{lr} = 2.5 \times 10^{-4}$, $\text{wd} = 1.0$; AdamW baseline $\text{lr} = 10^{-3}$, $\text{wd} = 3.0$. Selection protocol and full sweeps in Appendix B.

Group	Parameters	Optimizer
Hidden	QKV, attention output, MLP input, MLP output, per block	Muon
Auxiliary	Token embedding, position embedding	AdamW
Readout	Unembedding	AdamW

3 EARLIER GROKING IN THE TESTED MUON CONFIGURATIONS

The configuration-selection criteria differ. AdamW uses the fastest strictly stable configuration in its single-seed sweep; every tested Muon configuration is unstable, so Muon uses the fastest of those configurations (Appendix B). Across five seeds at these selected settings, median grokking steps are 8,300 for AdamW and 5,400 for Muon, a $1.54\times$ ratio. The fastest successful single-seed sweep entries reach the criterion at 6,300 and 5,400 steps, respectively, a $1.17\times$ ratio. These summarize different comparisons.

All nine entries in Muon’s 3×3 hidden-group sweep grok, compared with seven of eleven AdamW entries. Their success-conditional mean times are 13,011 and 30,486 steps, a $2.34\times$ ratio. The grids differ and this mean excludes four unsuccessful AdamW configurations. In the selected five-seed runs, Muon’s grokking steps span 100 steps and AdamW’s span 7,700. Ratios reach 4.8–4.9 under half width and a smaller modulus. AdamW never groks in the 20% training variant, yielding a horizon-dependent lower bound of $6.6\times$ (Appendix B); the depth comparisons are in §7.

All nine Muon configurations lose test accuracy after grokking; six have a logged evaluation with both train and test accuracy below 95%. None is strictly stable: sub-threshold counts range from 2 to 592, post-grokking minima reach 0.78%, and the counts have no monotone relationship to either hidden hyperparameter. Four of the seven grokking AdamW configurations also collapse, with severity rising monotonically in the learning rate, from none at 10^{-3} to 918 sub-threshold evaluations at 10^{-2} . Sweeping the readout learning rate over a factor of five and its weight decay from zero to 0.5 changes the timing and depth of collapse and eliminates none of it; the zero-decay case removes the weight-decay pull on the readout and leaves the failure in place. Across five seeds neither selected setting is strictly stable: Muon records 137 to 321 sub-threshold evaluations on every seed with minima to 16.27%, and the AdamW reference records one or two on four of five, with a minimum of 27.59%. The tested Muon subtraction run also loses test accuracy; its matched AdamW run stays above threshold. Full sweeps, the baseline-selection protocol, and generality variants are in Appendix B.

4 LOCALIZING THE COLLAPSE AT THE INTERFACE

4.1 MUON UPDATES REMAIN NEARLY CONSTANT AT SMALL GRADIENT NORMS

Figure 4 measures steps 44,010–44,700 of instrumented continuation I, sampled every update (Appendix A.1). Its first recorded training failure is at 44,702. Training loss is near 1.5×10^{-7} , with hidden and non-hidden gradient norms of order 10^{-6} and 10^{-5} . Muon normalizes its momentum before a fixed Newton–Schulz iteration, reducing sensitivity to its magnitude. Across this window, the log–log slope of the applied hidden step against the hidden gradient norm is -0.03 . Slopes for the embedding and readout steps against the combined non-hidden norm are about $+1.5$, with R^2 across the three fits ranging from 0.84 to 0.98. The hidden step and decay largely cancel; their net displacement grows by 64% from step 44,100 to 44,700 while the gradient-driven step changes little. By step 44,700, root-mean-square displacement per parameter from the branch state is 0.073 for hidden parameters and 0.0091 for non-hidden parameters, a factor of 8.0. These measurements establish different displacement scales; their causal interpretation requires the numerical controls in §4.4. In the separate ten-step replay used for Figure 3 and Table 11, training loss rises by 3.8×10^8 across one interval, and training and test accuracy fall from 100% to 21.12% and 19.04%.

4.2 WHICH PARAMETERS THE FAILURE REQUIRES

Table 2: Two branch suites from the reported step-44,000 source, each run to step 46,000 and evaluated every ten updates. The group and component suites have separate controls, G and C. Failure here means sampled training accuracy below 90%. Freezing either optimizer-managed side of the interface prevents it over this window; among non-hidden parameters only freezing the unembedding does, and hidden matrices together with the unembedding reproduce it.

Branch	First train collapse	Min test	Final test	Max non-hidden grad.
<i>Group branches</i>				
Control G	44,700	16.77%	100.00%	913.07
Freeze hidden	none	99.07%	99.08%	2.79×10^{-5}
Freeze non-hidden	none	100.00%	100.00%	0
<i>Non-hidden component branches</i>				
Control C	44,710	43.57%	99.93%	329.29
Freeze token embedding	44,910	34.74%	99.90%	3278.60
Freeze position embedding	44,760	37.78%	99.92%	323.76
Freeze unembedding	none	100.00%	100.00%	4.94×10^{-6}
Hidden and unembedding only	45,290	40.80%	100.00%	348.00

Table 2 reports two separately executed branch suites, each with its own control and 2,000-step horizon. Their controls have identical logged losses and accuracies at 44,000 and already differ at 44,010. Their earlier source trajectory and the limits of its retained checkpoint and runtime records are described in Appendix A.1. Freezing either group suppresses the failure for the whole window; freezing the token or position embedding delays it without preventing it; freezing the unembedding prevents it over the same window, and leaving only the hidden matrices and the unembedding free reproduces it. With the non-hidden parameters frozen, Muon continues to apply hidden updates near 0.146 while loss remains near 1.5×10^{-7} .

These branches establish suppression over a finite window; freezing also changes subsequent gradients and optimizer history. To isolate the failing update, Figure 1a swaps features and heads between adjacent checkpoints at five captured events. In every seed, the old head retains or improves accuracy on the updated features, while the new head sharply lowers accuracy on the old features. Fresh linear decoders fitted using only training examples recover 98.20–100% at selected collapsed checkpoints (Appendix J.2). The same protocol gives 0.04–0.23% at initialization and 0.02–1.12% at sustained memorization, supporting retention of information acquired during training; all selected refits converge (Appendix J.8). Longer branches still show coadaptation: each decodes its own addition-filtered representation and fails on the other’s, and initially identical heads reach cosine similarity -0.028 . These observations establish readout sensitivity and surviving information. Exact global basis equivalence fails for the measured pairs, despite successful recovery of planted transformations. The joint normalization/orthogonalization ablation also has a separate limitation: non-finite loss was observed in all six generalizing runs (Appendix D).

4.3 FREEZING EMBEDDINGS AND READOUT STABILIZES THE TESTED DEPTH-1 RUNS

We hold the token embedding, the position embedding, and the unembedding constant from roughly two thousand steps after sustained generalization, while hidden matrices continue to receive Muon updates; we call the configuration *Stable Muon*. Across five runs spanning the main condition, $p = 97$, training fraction 20%, and width 64, a total of 451,400 post-grokking steps and 4,519 evaluations, no evaluation falls below 95%, and in every run the post-grokking minimum is the accuracy at the evaluation that first met the criterion. Figure 2 pairs the arms at five seeds: freezing removes every sub-threshold evaluation on every seed, against an AdamW reference that achieves strict stability on one seed of five. Grokking speed is preserved because the freeze follows the threshold, and per-step cost falls afterward. In the long component comparison, holding only the unembedding fixed leaves eighteen of 936 evaluations below threshold; also freezing the embeddings removes them (Appendix E). Arithmetic correction also reaches the original 100,000-step budget (§4.4).

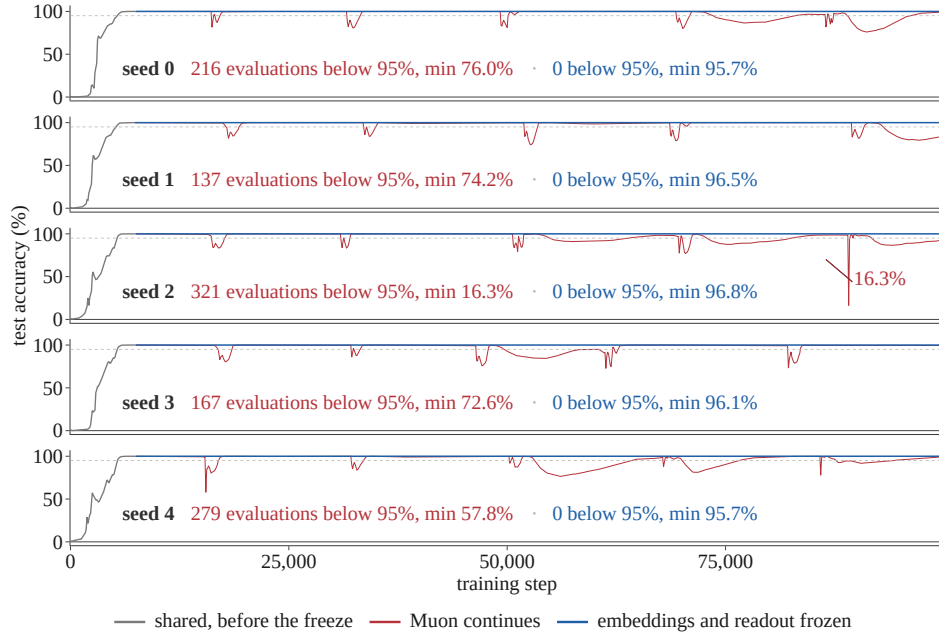


Figure 2: The intervention at five seeds. Before the freeze, the paired arms have identical logged metrics. One arm continues under ordinary Muon; the other holds embeddings and unembedding constant. The unfrozen arm falls below threshold 137 to 321 times per seed with minima as low as 16.27%; the frozen arm records none and never falls below 95.67%.

4.4 NUMERICAL CONTROLS AND THE FAILING STEP

Five matched branches replace stock cross-entropy with a cancellation-resistant `loglp` evaluation from step 6,000, after grokking and measurable derivative error. Through step 100,000, the original training horizon, none has a joint train/test failure below 90%; minimum measured test accuracy is 98.86%, and all finish at 100%. Every-update training checks trigger test evaluation below threshold, detecting every joint event. Brief test-only excursions can fall between grid evaluations. From step 15,000 to 20,000, accurate loss, target-derivative repair, and per-example zero-sum gradient projection each give 0/5 failures (Appendix J.3). Projection retains substantial derivative error while restoring the class-sum identity, connecting the result to numerical feature inflation (Liu et al., 2026).

From matched step-6,000 states, reducing Muon’s hidden learning rate from 0.03 to 0.003, with stock cross-entropy and trainable embeddings and readout, gives 5/5 joint failures by step 100,000 (Figure 1c). This particular tenfold reduction is insufficient to stabilize the five matched continuations. With unchanged weight-decay coefficients, it also reduces the hidden decoupled weight-decay step (Appendix J.7).

The mean-dependent head update reproduces the acute failure. Before the captured failures, full-grid feature-mean norms reach 10,374–12,203. Appendix J.5 provides descriptive timelines from the separate archived Muon and AdamW seed study. Let h_1 be the post-update residual, $W_{U,0}$ the preceding head, ΔW_U its actual displacement, and μ the training-set mean of h_1 . Starting from the rescued logits $W_{U,0}h_1$, the head-induced change decomposes exactly as

$$\Delta z = \Delta W_U(h_1 - \mu) + \Delta W_U\mu. \quad (1)$$

The same training-estimated μ is used for all test examples. Adding only $\Delta W_U\mu$ nearly reproduces each failure (Table 3); its test accuracy differs from the full update by at most 0.33 percentage points. The centered term retains 87.20–100% test accuracy and at least 99.09% training accuracy. This evaluation-only intervention establishes local sufficiency of the mean-dependent term, a class-dependent logit offset shared by all examples. Feature growth and optimizer history remain coupled before the failing update.

Table 3: Test accuracy (%) under the decomposition in Eq. 1. Each partial intervention adds one term to $W_{U,0}h_1$. The full update gives $W_{U,1}h_1$. All means use training examples only.

Seed	Preceding head	Mean term	Centered term	Full update
0	96.39	32.11	95.26	31.87
1	94.18	9.42	92.48	9.09
2	100.00	0.93	100.00	0.93
3	91.01	53.34	87.20	53.60
4	98.66	66.07	98.46	66.33

Replacing gradients only at the terminal update still fails in all five seeds. Each complete update has negative directional derivative under the accurate training loss, and scales 0.001 and 0.01 lower that loss; the full update overshoots (Appendix J.4).

The corrected subtraction run also remains free of joint failure through 100,000 steps. With gain-free RMS and accurate CE from initialization, all four prospective CPU seeds fail after grokking, at steps 14,271–15,789. Embedding-only updates reproduce joint failure in each; retaining the old embeddings with updated hidden matrices and readout gives 98.41–100% test accuracy. Pre-update derivative errors are below 9×10^{-8} relative. Fresh decoders reach 2.65–93.82% after failure, against 98.15–100% before it, with all selected refits converged. These results replicate the historical seed-0 embedding failure and show variable surviving linear accessibility (Appendix J.9).

5 TASK-ALIGNED FOURIER PROJECTIONS

We view each cached state as a function of the operand pair. An orthonormal two-dimensional Fourier transform over all p^2 pairs partitions its modes into six disjoint families: non-constant *addition* (k, k) , *subtraction* $(k, -k)$, *a-only*, *b-only*, the constant mode, and generic interaction. Family *sufficiency* retains that family’s modes; *ablation* removes them. Both interventions retain the constant mode when targeting a non-constant family. Unless stated otherwise, Fourier accuracies and solved-checkpoint selection ($\geq 95\%$) use the full operand grid. Appendix F specifies the interventions.

The filter supplies task structure. Retaining all (k, k) modes and the constant mode is exactly the orbit average

$$(P_{\text{add}}h)(a, b) = \frac{1}{p} \sum_{t=0}^{p-1} h(a+t, b-t), \quad (2)$$

with indices modulo p . Every donor has the same sum, and the linear readout commutes with this average. For subtraction, the shifts are $(a+t, b+t)$. A lookup-table predictor that outputs the correct one-hot vector on training examples and zero elsewhere becomes perfect after addition filtering whenever every answer orbit contains a training example. Appendix K verifies this control on the experimental splits. Full-family filtered accuracy thus measures the answer-conditioned average of the predictor; high accuracy alone leaves the learned algorithm unidentified.

Task-aligned projection preserves accuracy at the sampled solved checkpoints. In Table 4, the addition family reaches 100.00% at every selected addition checkpoint. Removing it leaves near-chance accuracy, while removing the subtraction family has little effect. Under matched initialization, configuration, and code path, training on $(a-b) \bmod 113$ exchanges these roles: $(k, -k)$ reaches 100% and (k, k) gives chance accuracy. These results characterize the task-aligned projection and family removal; both interventions retain the constant mode. The averaging identity also produces this operation dependence, so identifying shared learned algorithms requires additional evidence (Chughtai et al., 2023).

Frequency and phase matter within the family. Transporting the family onto other diagonal frequencies at matched power gives 2.64%. Randomizing phase per conjugate pair gives 0.19%, and randomizing phase per channel gives 0.78%. These interventions establish sensitivity to frequency assignment and phase. The memorizer controls in Appendix K also exhibit sparse filtered sufficiency and perturbation sensitivity, so these tests leave algorithm identification unresolved. Two branches

Table 4: Family sufficiency and ablation on the two operations. Addition columns at depths 2 and 4 aggregate eleven solved checkpoints; depth-1 columns are matched to the subtraction runs on initialization, configuration, and code path. Chance is $1/113 \approx 0.885\%$; across forty-three solved checkpoints spanning five seeds and three regimes, the task-aligned family gives exactly 100.00% at every one (Appendix F).

	Addition, depths 2 and 4	Addition, depth 1	Subtraction, depth 1
(k, k) sufficiency	100.00%	100.00%	0.88%
$(k, -k)$ sufficiency	0.88%	0.88%	100.00%
(k, k) ablation	1.34%	1.51%	99.68%
$(k, -k)$ ablation	98.26%	97.51%	2.54%
a -only / b -only / constant suff.	0.88%	0.88%	0.88%
Generic sufficiency	1.38%	1.59%	2.38%
Solved checkpoints	11	6	4

diverging from a bit-identical state each decode their own addition-filtered representation at 100% and the other’s at chance, showing dependence on the associated readout.

The update rule shapes the spread. On matched runs the Muon representation occupies 326 effective conjugate pairs of the non-constant spectrum against AdamW’s 5, leaving the task-aligned family holding 28% of non-constant power against AdamW’s 91%, and the contrast replicates across seeds and inside the family itself (Figure 5; Appendix G). These operand-frequency measurements complement prior work on Muon’s spectral properties (Ruan et al., 2026; Wang et al., 2025; Wang, 2026). Ablating the normalization and orthogonalization together collapses the count to 4, past AdamW’s, while task-family filtered accuracy remains 100%. These comparisons show a change in spectral dispersion. The filtered accuracy retains the averaging interpretation above.

6 INTERFERENCE AND SPECTRAL CHANGE

The task-aligned projection can fail with the model or remain accurate while the full model fails (Figure 11). At a masked checkpoint, full-grid filtered accuracy is 100% and full-model accuracy is 45.85%. Rescaling the non-constant task-aligned component raises accuracy to 99.91% without retraining. This is an intervention on a task-informed projection. Appendix L gives the margin decomposition, the scaling curve, and the early AdamW observation of 95.14% full-grid filtered accuracy alongside 0.30% held-out native accuracy. The memorizer control in §5 limits what these filtered accuracies establish.

In the ten-step spectral replay, dominant-frequency membership is unchanged across collapse and task-family power retains cosine similarity 0.9899 (Figure 3). The filtered margin nevertheless changes from +18.14 to −46.85. This supports the dissociation between power summaries and predictions. Single-update swaps provide the finer causal localization in Figure 1; exact basis tests and the full spectral analysis are in Appendices J.2 and L.

7 CONSTRUCTION AND ALIGNMENT SEPARATE IN DEPTH

At depth 2, Muon and AdamW reach the grokking criterion at 2,300 and 5,100 steps within a 100,000-step budget ($2.22\times$). At depth 4, Muon reaches it at 52,600; AdamW remains censored at 300,000, giving a lower bound of $5.7\times$. The AdamW setting selected for stability at depth 1 records 556 sub-threshold evaluations at depth 2. At depth 4, an exact-state freeze of embeddings and readout at step 126,200 reduces sub-90% evaluations from 1,353 to 66 and raises mean post-branch accuracy from 48.07% to 97.50% through step 300,000 (1,738 post-branch evaluations at 100-step intervals; Appendix I).

Across 41 solved checkpoints, intermediate post-block residuals decoded by the final head never exceed 17.97%, and only the final block reaches 95%. This measures compatibility with that particular head. Appendix I retains the stagewise perturbation analyses, their five-perturbation averages, and the complete depth results.

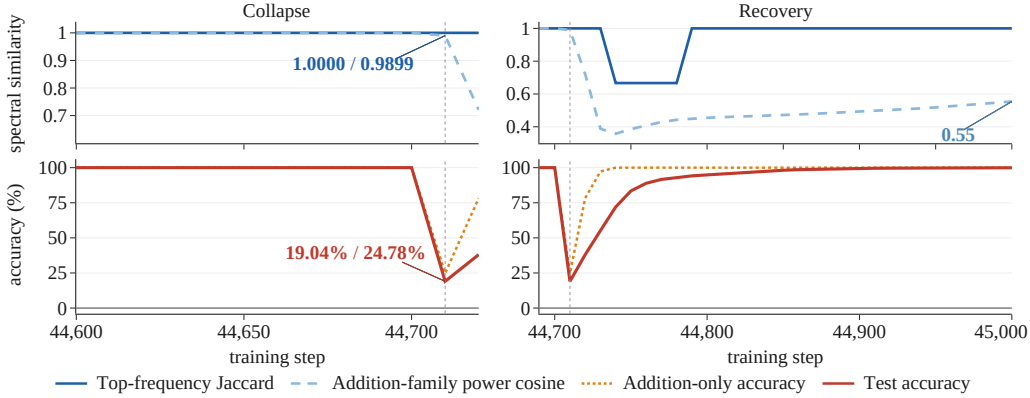


Figure 3: Spectral continuation S , evaluated every ten updates through step 45,000. The last sampled healthy state is 44,700 and the first sampled failure is 44,710 (Appendix A.1). **Left:** test accuracy falls from 100% to 19.04% and addition-filtered accuracy to 24.78%. The five dominant addition-frequency pairs are unchanged (Jaccard 1.0000), and power over all 56 addition pairs retains cosine similarity 0.9899. **Right:** accuracy returns to 99.90% within 300 steps, with power cosine 0.55. The same five pairs remain dominant, with a different allocation of power. The addition family consists of the non-constant (k, k) modes of the final residual’s operand-grid Fourier transform; accuracy filters also retain the constant mode.

8 RELATED WORK

Grokking on modular arithmetic. Power et al. (2022) introduced the phenomenon. Nanda et al. (2023) reverse-engineered a Fourier algorithm, used causal ablations, and introduced restricted and excluded losses to track circuit formation and cleanup. Zhong et al. (2023) show that hyperparameters select qualitatively different algorithms; Chughtai et al. (2023) find shared algorithmic structure alongside variation across learned circuits.

Numerical instability and adaptive updates. Thilak et al. (2022) report cyclic instability under adaptive optimizers; Varma et al. (2023) describe ungrokking under reduced data. Prieto et al. (2025) identify softmax collapse and a loss-minimization direction that scales logits. Liu et al. (2026) connect cross-entropy class-sum errors to feature/classifier-mean growth and amplified Adam updates, and test corrections to this numerical feedback. Our matched arithmetic branches extend this evidence to Muon/AdamW routing. Adjacent swaps localize individual failures, and intervention on the mean-dependent part of the actual readout displacement tests its local sufficiency. Joint train/test failure also differs operationally from the perfect-training regime of anti-grokking (Prakash & Martin, 2025).

Wortsman et al. (2024) reproduce attention-logit growth and output-logit divergence in small transformers at high learning rates, and show that independently scaling learning rate and weight decay reduces learning-rate sensitivity. This motivates treating our ordinary LR reduction as a joint change in update size and decay. Molybog et al. (2023) analyze Adam updates with growing magnitude and weak alignment to descent directions in large-model training. Our captured updates have negative accurate-loss directional derivatives but overshoot at their full scale, providing a measured finite-step qualification for these events.

Muon. Jordan et al. (2024) introduced the optimizer and Liu et al. (2025) deployed it at scale under the routing used here. Tveit et al. (2025) and Wang (2026) report faster grokking in their tested settings; the latter studies orthogonalization and the speed/stability tradeoff with iteration count, including accuracy dips. We examine the five-iteration configuration over longer horizons. Ruan et al. (2026) and Wang et al. (2025) report higher effective rank and a more isotropic weight spectrum under Muon. Our Fourier measurements concern dispersion over operand frequencies, and the joint ablation tests the contribution of normalization and orthogonalization.

Representation and readout. Elhage et al. (2021) describe residual-stream basis freedom. Anthes et al. (2023) identify readout misalignment as a major component of forgetting under new tasks. Chou et al. (2026) use same-split linear probes, geometry, and readout-learning diagnostics to separate representation quality from classifier calibration during grokking and double descent. Gomezjurado Gonzalez (2026) finds a decoder bottleneck in one-step Collatz prediction. Here the task stays fixed: probes at captured post-grokking failures are combined with adjacent-update swaps and a measured-displacement intervention. Initialization and memorization controls calibrate how much linear accessibility precedes the failure.

9 DISCUSSION

Training-only decoders reach 98.20–100%, while exact global basis equivalence fails for the measured pairs. Captured updates overshoot accurate local descent directions. Adjacent swaps, the feature-mean intervention, and arithmetic corrections support a numerical contribution to acute readout failure. The relative contributions of feature growth and optimizer history remain unresolved.

The accurate-loss RMS cohort exhibits embedding-localized failures and variable failed-state decodability. The unnormalized controls cover five selected trajectories; the operation extension uses one seed, and the RMS study adds four prospective seeds to a separately identified historical pilot. Behavior beyond 100,000 steps and brief test-only excursions remain unresolved; AdamW collapses limit Muon-specific attribution. Full-family Fourier filtering supplies answer-orbit structure and mixes training and test inputs. The memorizer control rules out inferring an arithmetic algorithm from perfect filtered accuracy alone. Accuracy can collapse through a damaging readout update while useful task information survives, and corrected arithmetic stabilizes the tested unnormalized runs through step 100,000.

REPRODUCIBILITY STATEMENT

The original experiments run on a single device from released code. The public repository at <https://github.com/Na00s/muon-grokking> contains the training and analysis scripts, 191 CSV run logs, and 20 HTML summaries. Follow-up artifacts add raw diagnostics, model/optimizer/random-state checkpoints, replay checks, and the scripts for every new table and figure. Matched freezes and arithmetic branches verify shared states before intervention. Appendix A specifies the original setup; Appendix J gives the decoder, basis, arithmetic, and architecture-control protocols and their fixed horizons.

AI USE STATEMENT

Generative AI tools were used for code completion suggestions and manuscript proofreading. Any generated content was verified, and the authors take full responsibility for the work and the final manuscript.

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A SETUP DETAILS AND OPERATIONAL DEFINITIONS

Table 5: Operational definitions. The original training grid is 100 updates. Five consecutive scheduled evaluations span 400 updates; follow-up confirmation uses six and spans 500. Finer event-capture schedules are identified by trajectory.

Term	Definition
Sustained 99.9% train	First of five consecutive scheduled training evaluations at or above 99.9%.
Original grokking	First of five consecutive scheduled test evaluations at or above 95%. Used for the original speed comparisons.
Memorization plateau	Interval from sustained-99.9% training accuracy to original grokking; its duration is the difference between these steps.
Follow-up confirmation	Sixth consecutive scheduled test evaluation at or above 95%. Prospective joint-event monitoring begins after this confirmation.
Strict stability	Every subsequent sampled test accuracy is at least 95% over the stated horizon.
Joint failure	Training and held-out test accuracy are both below 90% at the same state, after confirmation where required by the protocol.
Table 2 failure	Training accuracy below 90% at a scheduled branch evaluation. This criterion does not require a simultaneous test threshold.
First detected failure	Earliest qualifying observation under the stated monitoring schedule. It can follow the underlying threshold crossing when sampling is sparse.
First captured failure	Earliest qualifying saved state under the specified capture protocol. Its window and resolution can differ from the original detection schedule.
Minimum sampled accuracy	Minimum over recorded evaluations in the stated window, with the inclusion of the source or confirmation state specified.
Final accuracy	Accuracy at the reported terminal state; recovery can make it higher than the post-grokking minimum.
Evaluation count	Number of recorded evaluations satisfying the stated threshold, distinguishing scheduled and training-triggered evaluations.
Seed-level incidence	Number of seeds with at least one qualifying event divided by the number of evaluable seeds over the stated horizons. Repeated events within a seed count once.

Each example is the token sequence $[a, b, =]$; the vocabulary contains $p + 1 = 114$ tokens and the output layer $p = 113$ classes. The ordered operand grid contains $p^2 = 12,769$ examples; a fixed random 30% (unless stated) forms the training set, generated once from the run seed and shared by all matched comparisons. The original sweeps load saved initial states, evaluate every 100 steps, and checkpoint every 1,000. Their budgets are 100,000 steps at depths 1 and 2 and 300,000 at depth 4. Diagnostic branches use the shorter horizons and finer grids specified below.

Each block holds four two-dimensional matrices (fused QKV 384×128 , attention output 128×128 , MLP input 512×128 , MLP output 128×512), 196,608 parameters; with embeddings and unembedding the depth-1 model has 226,048 parameters. Depth is varied by appending blocks under nested initialization, so deeper models begin from a strict superset of the shallower model’s initial state. Muon maintains an exponential moving average of the gradient with $\beta = 0.95$, forms a Nesterov-style update, divides it by its Frobenius norm plus 10^{-7} , applies five quintic Newton–Schulz iterations with coefficients $(3.4445, -4.7750, 2.0315)$, scales by $\sqrt{\max(1, \text{rows}/\text{cols})}$, and applies decoupled weight decay; the implementation refuses any parameter that is not a matrix. All AdamW groups use $\beta_1 = 0.9$, $\beta_2 = 0.999$.

Table 5 distinguishes the original and follow-up definitions. Original scheduled evaluations occur every 100 steps; their five-check criterion spans 400 steps, while follow-up confirmation occurs on the sixth qualifying check. The original study reports accelerator execution for the single-seed sweeps and CPU execution for the five-seed replications. Historical runtime manifests are incomplete (Appendix A.1). The Muon and Stable Muon CPU arms have identical logged metrics over the seventy-four to seventy-five evaluations preceding the freeze. This check compares recorded metrics; it does not compare every parameter tensor at each evaluation. Fresh follow-ups use the explicitly recorded deterministic CPU configuration and verify the replays described in Appendix J. The seed sets the initialization and the train/test split together, so the five seeds are five draws of the problem instance.

Table 6: Selected configurations. Stable Muon and Muon are the same configuration up to the freeze; their pre-freeze trajectories can differ through backend nondeterminism, which Section 4.3 reports.

Configuration	Group	Optimizer	Settings
AdamW baseline	All parameters	AdamW	$\text{lr} = 10^{-3}$, $\text{wd} = 3.0$
Muon	Hidden	Muon	$\text{lr} = 0.03$, $\text{wd} = 0.1$, $\beta = 0.95$, $\text{NS} = 5$
	Auxiliary	AdamW	$\text{lr} = 10^{-3}$, $\text{wd} = 1.0$
	Readout	AdamW	$\text{lr} = 2.5 \times 10^{-4}$, $\text{wd} = 1.0$
Stable Muon	Muon, with auxiliary and readout groups frozen after sustained generalization		

A.1 HISTORICAL CONTROLS AND EVENT IDENTITIES

Table 2 contains two separately executed control continuations. Control G belongs to the parameter-group suite, and Control C to the auxiliary-component suite. Both report the same losses and accuracies at step 44,000, matching the archived hidden-LR-0.01, seed-0 trajectory: training loss $1.86781946 \times 10^{-7}$, test loss $3.32467862 \times 10^{-5}$, and 100% accuracy on both splits. Their logged losses already differ at 44,010. Each suite is evaluated every ten updates through 46,000, giving 201 samples including the source state. The reported minimum test accuracies, 16.77% and 43.57%, reproduce the corresponding CSV minima; final accuracies and maximum auxiliary-gradient norms likewise reproduce the last row and column maxima. The first-failure column uses training accuracy below 90% after memorization.

The historical branch programs load the model, Muon, and auxiliary AdamW state dictionaries. This earlier design uses one AdamW group for embeddings and readout. They reseed Python and PyTorch from the saved seed, including CUDA when available; their checkpoint format omits saved RNG states and backend metadata. The model is full-batch and has no dropout. The original step-44,000 checkpoint tensors, launch commands, and runtime manifests are unavailable in the retained artifacts. Matching initial metrics therefore cannot certify bitwise equality of the two restored states or identify the original backend. The logged results remain verified; the source of their subsequent divergence is unresolved.

We separately test implementation parity on an available seed-0, step-16,000 checkpoint. To match the earlier optimizer layout, we merge the head into the auxiliary AdamW group at learning rate 0.001, preserving all saved moments and step counters. The historical group, component, and instrumented group programs each run 2,000 updates on CPU with PyTorch 2.13.0, one thread, and deterministic algorithms. All 201 shared metric rows agree exactly, and terminal model, optimizer, and PyTorch RNG states are bitwise equal. This replay verifies agreement of the three implementations under the registered CPU test; its adapted source state is separately identified in the artifacts.

Table 7: Distinct historical continuations around step 44,700. Accuracies are percentages at each run’s first observed training failure. The spectral comparison uses S’s healthy 44,700 state and failing 44,710 state.

ID	Analysis	Sample interval	End step	Failure step	Train / test
G	Table 2, groups	10	46,000	44,700	18.54/16.77
C	Table 2, components	10	46,000	44,710	47.70/43.57
I	Applied-step diagnostics	1	46,000	44,702	24.31/22.12
S	Spectral replay	10	45,000	44,710	21.12/19.04

The source files under `runs/` are `branch_control_from_44000.csv` (G), `aux_control_from_44000.csv` (C), `branch_control_from_44000_instrumented.csv` (I), and `collapse_spectral_replay_from_44000.csv` (S). Their sampling intervals and terminal horizons identify separate trajectories even when event timestamps coincide. The ten-update grids can miss intervening excursions. Source hashes, exact summary calculations, and the controlled replay protocol accompany the paper.

Table 8: AdamW sweep. Seven of eleven retained configurations reach sustained 95% test accuracy. Post-grokking evaluations below 95% increase monotonically with the learning rate.

LR	WD	Sustained 95%	Min after grokking	Evals < 95%	Strictly stable
3×10^{-4}	0.1	no grok			
3×10^{-4}	1.0	no grok			
3×10^{-4}	3.0	25,500	95.67%	0	yes
10^{-3}	0.1	no grok			
10^{-3}	1.0	52,500	96.49%	0	yes
10^{-3}	3.0	8,200	95.01%	0	yes
3×10^{-3}	0.1	49,500	13.26%	2	no
3×10^{-3}	1.0	8,500	53.72%	4	no
10^{-2}	0.1	62,900	0.54%	258	no
10^{-2}	1.0	6,300	0.41%	918	no
10^{-2}	3.0	no grok			

Table 9: Muon sweep over the hidden group. All nine configurations reach sustained 95% test accuracy and none is strictly stable. The count of sub-threshold evaluations has no monotone relationship to either hyperparameter.

LR	WD	Sustained 95%	Min after grokking	Evals < 95%	Strictly stable
0.03	0.1	5,400	69.40%	56	no
0.01	0.1	6,000	53.24%	107	no
0.03	0.3	7,200	75.63%	2	no
0.01	0.3	8,500	76.60%	145	no
0.03	0.03	9,000	0.78%	592	no
0.003	0.1	14,400	90.07%	5	no
0.003	0.03	19,000	21.24%	130	no
0.01	0.03	19,500	0.83%	473	no
0.003	0.3	28,100	78.59%	265	no

B SWEEPS, BASELINE SELECTION, AND GENERALITY

Tables 8 and 9 give both sweeps under identical architecture, split, and budget. The fastest AdamW configuration ($\text{lr} = 10^{-2}$, $\text{wd} = 1.0$) is unusable: 918 sub-threshold evaluations at this seed, and across seeds three of five never reach sustained 95%, the two that do record 805 and 956, and all five end near chance. The baseline is therefore the fastest configuration that is strictly stable in this sweep, $\text{lr} = 10^{-3}$, $\text{wd} = 3.0$; the seed replication shows strict stability at that setting is a property of one seed rather than of the configuration. For Muon no configuration is strictly stable, so the selected configuration is the fastest.

Both optimizers fit the training set between steps 200 and 800 in every grokking configuration, so the difference lies almost entirely in the delay from memorization to generalization: 30,100 steps for AdamW against 12,544 for Muon averaged over successes. Across five seeds Muon reaches sustained generalization between 5,300 and 5,400 steps and the AdamW reference between 4,000 and 11,700, medians 5,400 and 8,300.

Varying the readout learning rate over 10^{-4} , 2.5×10^{-4} , and 5×10^{-4} gives post-grokking minima of 37.13%, 79.40%, and 3.81%; varying its weight decay over 0, 0.1, and 0.5 gives 33.34%, 0.79%, and 1.12%; every setting is unstable. On $(a - b) \bmod 113$ the Muon run records 18 post-grokking evaluations below 95% with a minimum of 1.52% and the AdamW run records none, reproducing the depth-1 addition asymmetry at 194 and none. Table 10 varies modulus, training fraction, and width; Stable Muon and Muon share a configuration up to the freeze, so differences in their grokking steps bound run-to-run variation on this backend.

Table 10: Controlled variation of modulus, training fraction, and width. Stable Muon and Muon share a configuration up to the freeze, so differences in their grokking step measure pre-intervention run-to-run variation on this backend.

Variant	Regime	Sustained 95%	Min post-grokking	Outcome
$p = 97$, train 30%, width 128	AdamW	31,000	95.52%	stable
	Muon	6,300	15.49%	unstable
	Stable Muon	6,300	95.05%	stable
$p = 113$, train 20%, width 128	AdamW	no grok		
	Muon	15,100	49.25%	unstable
	Stable Muon	14,500	95.16%	stable
$p = 113$, train 30%, width 64	AdamW	84,800	96.38%	stable
	Muon	17,700	53.23%	unstable
	Stable Muon	16,500	95.67%	stable

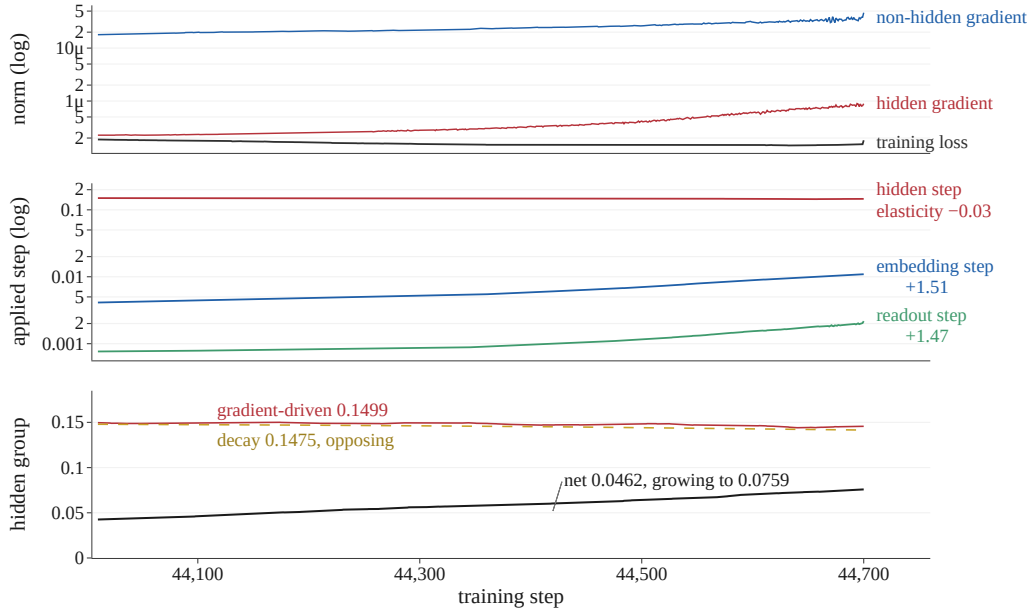


Figure 4: Every-step measurements over 691 updates before a collapse. **Top:** loss and gradient norms in the quiet window. **Middle:** gradient-driven applied steps and log-log slopes against the hidden gradient norm for Muon and the combined non-hidden norm for both AdamW groups. **Bottom:** hidden step and decay magnitudes differ by under 2% at step 44,100. Their vector sum is smaller and grows by 64% from step 44,100 to 44,700 while the gradient-driven step changes little.

C THE QUIET WINDOW

Figure 4 uses the every-step instrumented rerun, with a quiet window from 44,010 through 44,700 and failure at 44,702. Table 11 uses a separate ten-step spectral replay, whose first observed failure is at 44,710. The following step/decay diagnostics use the instrumented rerun. On the hidden group the gradient-driven component is 0.1499 at step 44,100 against a decay component of 0.1475, a ratio of 0.984, leaving a net displacement of 0.0462 that grows to 0.0759 across the window while the orthogonalized step stays flat; over the window the hidden gradient norm grows by 3.92 while its gradient-driven step changes by 0.97, and the non-hidden gradient grows by 2.60 while the embedding and readout steps grow by 2.65 and 2.84. The readout decay component has larger norm than its gradient-driven component on 14.9% of quiet-window steps. In the ten-step spectral replay, the combined non-hidden gradient norm rises from 3.52×10^{-5} to 836.10; the orthogonalized

Table 11: Diagnostics through the quiet window and across the collapse. The training loss and both gradient norms are numerically negligible until the failure. The Muon update column is the orthogonalized step before weight decay, varying by under three percent across the pre-collapse rows. These rows come from spectral continuation S, sampled every ten updates. Net-step diagnostics use continuation I, sampled every update, whose first training failure is 44,702 (Appendix A.1).

Step	Train loss	Hidden grad. norm	Non-hidden grad. norm	Muon update norm
44,100	1.77×10^{-7}	2.33×10^{-7}	1.95×10^{-5}	0.1498
44,300	1.55×10^{-7}	2.73×10^{-7}	2.20×10^{-5}	0.1494
44,500	1.50×10^{-7}	4.06×10^{-7}	2.67×10^{-5}	0.1481
44,700	1.53×10^{-7}	8.64×10^{-7}	3.52×10^{-5}	0.1455
44,710	57.50	25.26	836.10	0.1429
44,720	6.58	5.09	92.55	0.1655

Table 12: Ablated runs that reach sustained generalization. Minimum is the lowest test accuracy from the sustained-95 step to the last recorded evaluation. Terminal is the step at which non-finite loss was observed during the original run.

Learning rate	Backend	Precision	Grokked	Minimum	Terminal
0.3	MPS	float32	3,100	97.28%	23,514
0.3	CPU	float32	3,100	97.52%	24,485
0.3	CPU	float64	3,100	97.06%	22,889
1.0	MPS	float32	6,800	97.11%	23,800
1.0	CPU	float64	7,700	95.96%	29,831
1.0	CPU	float32	6,500	95.60%	38,955

Muon step over that interval is 0.1429, lower than the 0.1455 preceding it, and its maximum over the branch occurs after the event.

D ABLATING THE NORMALIZED-ORTHOGONALIZATION PATH

The ablation removes Frobenius normalization and the Newton–Schulz iteration together, retaining momentum, shape scaling, and weight decay. In the tested sweep, learning requires a rate of 0.3 or 1.0. Six ablated runs generalize, with no recorded post-grokking evaluation below 95% (Table 12). All six were observed to terminate with non-finite loss, in float32 and float64, at steps 22,889–38,955. The matched Muon comparator records thirteen sub-threshold evaluations through step 22,800. Over its full 100,000 steps, its minimum and final accuracies are 29.49% and 96.48%. At the analyzed checkpoints, ablation reduces dispersion from 326.09 effective conjugate pairs to 4.11, while the task-aligned family retains 100% sufficiency (Appendix G).

E PREVENTION DETAILS

In four of the five runs the freeze is triggered automatically, 2,000 steps after the first evaluation of a five-evaluation streak at or above 95%; the main-condition run uses a fixed step. The first logged frozen evaluation lands 2,100 to 2,200 steps after sustained generalization; the actual freezes occur 100 steps earlier. Within a depth-2 run the median step costs 57.87 ms before the freeze and 54.72 after; at depth 4, 111.39 and 107.75; pre-freeze rates match ordinary Muon’s. Holding only the unembedding constant from the same step leaves the run unstable, minimum 83.13% and eighteen of 936 evaluations below 95%; freezing the embeddings and unembedding together at the same step produces none.

F FOURIER DECOMPOSITION AND INTERVENTION SUITE

For a chosen point in the forward pass the model runs on all p^2 operand pairs, the states form a tensor $X \in \mathbb{R}^{p \times p \times d}$, and an orthonormal two-dimensional DFT over the operand axes gives $\hat{X}[k, l, :]$ with

Table 13: Stable Muon across five runs covering four settings; the main condition appears twice, under a fixed freeze step and under the automatic trigger. The minimum is the lowest test accuracy at any evaluation from the sustained-95 step onward, and in every run it is the value at that step itself.

Run	Grokked	First frozen eval.	Post-grokking steps	Minimum	Evals < 95%
Main condition	5,900	8,100	94,100	95.34%	0
Selected Stable Muon	5,400	7,500	94,600	95.74%	0
$p = 97$	6,300	8,400	93,700	95.05%	0
Training fraction 20%	14,500	16,600	85,500	95.16%	0
Width 64	16,500	18,600	83,500	95.67%	0
Total			451,400		0

Table 14: The six mode families. The families are pairwise disjoint and exhaust the $p \times p$ grid, so any statement about one family and its complement is an exact accounting of the whole representation.

Family	Modes	Count at $p = 113$
Addition	$(k, k), k \neq 0$	112
Subtraction	$(k, -k), k \neq 0$	112
a -only	$(k, 0), k \neq 0$	112
b -only	$(0, l), l \neq 0$	112
Constant	$(0, 0)$	1
Generic interaction	all remaining (k, l)	12,320
Total		12,769

power $\|\hat{X}[k, l, :]\|^2$. Because X is real, modes occur in conjugate pairs and every filter retains or removes both members together; transforms run in single precision on CPU. The families of Table 14 are pairwise disjoint and exhaust the grid, and addition and subtraction are disjoint because p is odd. Power fractions are normalized against non-constant power; every non-constant-family sufficiency filter also retains the constant mode, which becomes a fixed logit offset under the bias-free unembedding. Ablation removes only the selected non-constant modes, retaining this offset. The two masks are complementary on the non-constant spectrum and share the constant mode. In the margin decomposition and α sweep, the task-aligned component excludes the constant mode, which belongs to the remainder. Solved-checkpoint selection and intervention accuracies use the full operand grid; training trajectories explicitly labeled test accuracy use the held-out split.

Table 16 applies sufficiency and ablation to each family at forty-three solved checkpoints spanning five seeds and the Muon, Stable Muon, and AdamW regimes. Equal-power relocation ranges 0.88% to 15.04% over 430 replicates. Among the tested descending-power prefixes, the smallest reaching 95% contains two to four conjugate pairs at AdamW checkpoints (median 3), against three to seventeen at Muon (median 12.5) and eight to nineteen at Stable Muon (median 11). Along the AdamW trajectory of §6, the five strongest pairs give 4.91% at step 3,000, 19.48% at 20,000, 70.73% at 30,000, and 100% at 37,000, while the family’s share rises from 1.05% through 1.66% and 11.08% to 70.62%, reaching 90.51% by step 100,000; test accuracy over those steps is 0.30%, 0.29%, 2.08%, and 100%.

The full-family accuracies above include the orbit average in Eq. 2. Prefix filtering and phase or frequency perturbations measure additional spectral structure, but their interpretation as evidence of a learned algorithm requires controls against memorization. Appendix K supplies that comparison.

G SPECTRAL DISPERSION

Table 17 reports the entropy effective count $\exp(-\sum_i q_i \log q_i)$ over all 6,384 non-constant conjugate pairs and the task-aligned family’s power share on matched depth-1 runs. Here q_i denotes normalized pair power. Across the five-seed study AdamW occupies 4.40 effective pairs on average over thirty solved checkpoints, Muon 385.85 over twenty-five, and Stable Muon 343.96 over

Table 15: Interventions applied to a cached state. Preserved and altered quantities specify the intervention; an accuracy change measures sensitivity to that perturbation. Non-constant-family sufficiency and ablation both retain the constant mode.

Intervention	Preserved	Destroyed
Family sufficiency	Selected family and constant mode	Other non-constant modes
Family ablation	All other families	Modes in one family
Equal-power relocation	Total power, family membership	Which frequencies carry it
Random native subset	Pair count, native coefficients	Which of the used frequencies survive
Pair-global phase scramble	Support and power exactly	Relative phase across pairs
Channelwise phase scramble	Support and power exactly	Relative phase across channels
Cross-readout substitution	Both representations intact	Pairing of state with its own readout

Table 16: Family interventions at forty-three solved checkpoints, spanning five seeds and the Muon, Stable Muon, and AdamW regimes. Sufficiency retains the named family and constant mode; ablation removes the named non-constant family and retains the rest. Accuracies and selection use the full grid. The chance rate is 0.88%.

Family	Sufficiency	Range	Ablation	Range
Addition	100.00%	[100.00, 100.00]	1.71%	[0.67, 4.24]
Subtraction	0.88%	[0.88, 0.88]	99.61%	[95.79, 100.00]
<i>a</i> -only	0.88%	[0.88, 0.88]	99.85%	[97.76, 100.00]
<i>b</i> -only	0.88%	[0.88, 0.88]	99.85%	[97.72, 100.00]
Constant	0.88%	[0.88, 0.88]		
Generic interaction	3.28%	[0.78, 14.21]	99.99%	[99.78, 100.00]

thirty-six: freezing the interface leaves the dispersion where Muon put it and removes the instability anyway. Within each family’s $(p - 1)/2$ pairs, the generality analysis instead uses the inverse participation ratio $1/\sum_i q_i^2$, with q_i renormalized within the family: 1.99–2.59 for solved AdamW variants and 21.33–55.34 for Muon. On the uniform operand grid, a predictor using only the opposite family and a constant offset has accuracy exactly $1/p$, since sum and difference are independent for odd p . Across solved matched checkpoints, AdamW places 0.0–7.8% of non-constant power there and Muon 2.5–17.7%.

H THE MATCHED BRANCH THROUGH FAILURE, RECOVERY, AND MASKING

Table 18 follows the family’s top-twenty membership, power share, isolated accuracy, and full-model accuracy along the depth-4 matched pair of \$6 from a bit-identical state. Over 174,000 steps, the frozen branch retains high filtered accuracy and substantial family power; the control loses all three at filtered-pathway failure through its current readout, recovers all three, and loses the share again at projection masking while the isolated family still gives 100%. The α -sweep crossing at $\alpha = 2.55$ occurs at a share of 0.6247, below the frozen branch’s native 0.9069. Replaying a depth-1 collapse and recovery at ten-step resolution: full accuracy returns to 99.90% within 300 steps while the addition-family power cosine to the pre-collapse state reaches only 0.55; the pre-collapse representation holds between 4.2% and 5.7% under the recovered readout throughout, while the recovered representation reads at 90.73% under the pre-collapse readout. The matched branches’ five strongest modes are identical at the branch point and share at most one pair at sampled steps 160,000 through 230,000. Patching the addition component of the frozen branch’s final MLP write into the control under the frozen readout gives 99.99%; under the control’s own readout, 5.05%; patching its full final MLP write gives 97.61% and 4.46% under the two readouts; patching only the non-addition component of that MLP write into the control gives 36.79%, below the control’s unmodified 45.85%.

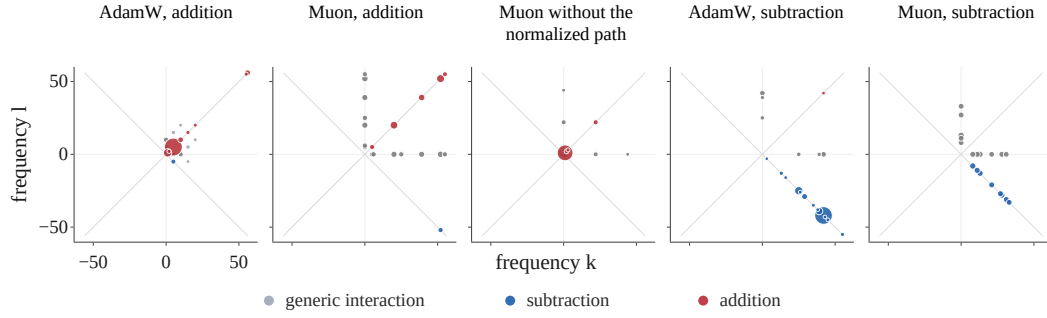


Figure 5: The twenty highest-power conjugate pairs of the final residual at converged checkpoints, on the mode grid, sized by share of non-constant power; grey lines mark $l = k$ and $l = -k$. AdamW places 80.1% of non-constant power on a single pair of the family the task selects; Muon’s strongest pair carries 7.9%, with the rest spread across the family and beyond it; ablating the normalized path returns the strongest pair to 58.9%. Between the addition and subtraction panels the occupied line moves from one diagonal to the other.

Table 17: Spectral dispersion at solved checkpoints of the matched depth-1 runs. Effective pairs is $\exp(-\sum_i q_i \log q_i)$, with normalized pair powers q_i , over all 6,384 non-constant conjugate pairs, reported as a mean with range. Family power is the task-aligned family’s share of non-constant power, meaning (k, k) for addition and $(k, -k)$ for subtraction, and top-five power is the share carried by the five strongest pairs of the spectrum. The final row ablates the normalization and orthogonalization together.

Operation	Optimizer	Effective pairs	Family power	Top-five power
Addition	AdamW	4.95 [3.68, 10.20]	91.0%	93.7%
Addition	Muon	326.09 [96.91, 578.90]	28.0%	13.1%
Subtraction	AdamW	2.54 [2.45, 2.72]	95.4%	96.5%
Subtraction	Muon	211.46 [33.25, 377.24]	27.7%	16.0%
Addition	Muon, path ablated	4.11 [2.04, 6.05]	66.5%	94.0%

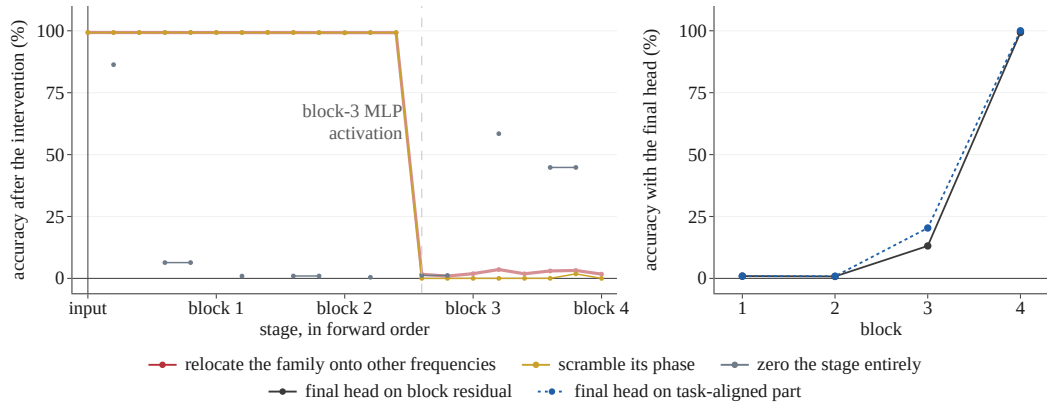


Figure 6: Depth-4 Muon at step 53,000. **Left:** the reported stage-intervention realization. Relocation and phase scrambling first cause a large drop at the block-3 MLP activation. The two stochastic curves use replicate 4 of five and illustrate sensitivity; individual later-stage outcomes vary across perturbations. Zeroing the earlier block-2 and block-3 attention writes also destroys accuracy, despite their low sensitivity to these Fourier perturbations. **Right:** direct decoding through the final unembedding, using each post-block residual or its addition-filtered component. Both reach high accuracy only at the final block.

Table 18: The addition family along matched branches. Pairs counts how many of the twenty highest-power conjugate pairs at the final residual belong to the family; share is the family’s fraction of non-constant power; addition-only is accuracy when the representation is filtered to the family.

Step	Branch	Pairs	Share	Addition-only	Full
126,200	control	11	83.95%	100.00%	94.07%
126,200	freeze	11	83.95%	100.00%	94.07%
160,000	control	1	11.88%	2.65%	3.29%
160,000	freeze	12	89.33%	100.00%	98.97%
180,000	control	0	0.32%	0.88%	0.97%
180,000	freeze	12	88.32%	100.00%	97.95%
230,000	control	4	67.69%	100.00%	98.41%
230,000	freeze	12	90.78%	100.00%	99.47%
300,000	control	3	20.38%	100.00%	45.85%
300,000	freeze	12	90.69%	100.00%	98.89%

Table 19: Time to sustained generalization, in optimizer steps and in elapsed seconds on a single device. Per-step cost is the median over evaluation intervals. The AdamW baseline at depth 4 is right-censored at the step budget.

Depth	Optimizer	ms/step	Steps to grok	Seconds to grok	Advantage
2	AdamW	33.06	5,100	170	2.22× steps, 1.28× time
	Muon	57.81	2,300	133	
4	AdamW	61.68	> 300,000	> 19,768	> 5.70× steps, > 3.37× time
	Muon	110.87	52,600	5,856	

I DEPTH

Muon reaches sustained generalization in 2,300 steps at depth 2 against 5,100, and in 52,600 at depth 4 where AdamW peaks at 96.69% and never sustains the threshold within 300,000 steps. Table 19 compares steps and elapsed time; a Muon step costs 1.75–1.80× an AdamW step. At the depth-4 checkpoint in Table 20, zeroing the block-2 and block-3 attention writes leaves 0.94% and 0.44% accuracy. Frequency relocation and phase scrambling at these writes preserve near-baseline accuracy. Both perturbations cause a large drop at the block-3 MLP activation. Figure 6 and Table 20 show the reported perturbation realization. Later-stage outcomes vary across the five perturbations, so near-chance accuracy in this realization does not establish uniform sensitivity across perturbations.

Across eleven solved checkpoints, the first stage where mean pair-phase scrambling costs more than half the baseline accuracy is an MLP activation in ten and an MLP write in one. This criterion averages five perturbations per stage. Decoding intermediate residuals with the model’s final head measures compatibility with that particular head. Under this head, accuracy reaches 95% only at the final block across all 41 checkpoints selected by full-grid accuracy, never exceeding 17.97% earlier (Table 21). Within the eleven-checkpoint stage-intervention subset, among seventeen residual states with direct-readout accuracy between 2% and 95%, family filtering improves sixteen, with a mean increase of 4.88 percentage points across all seventeen. Across eighteen depth-4 checkpoints after the shared state, between-branch power cosine stays above 0.98 in the first block; in the fourth it lies between 0.03 and 0.44 at fifteen checkpoints. The exact-state branch at step 126,200 raises mean post-branch accuracy from 48.07% to 97.50%, final accuracy from 36.88% to 98.87%, and cuts evaluations below 90% from 1,353 to 66.

J DECODER, BASIS, AND ARITHMETIC FOLLOW-UPS

The decoder and matched-branch tests use the original architecture, data splits, and optimizer settings; operation and normalization controls are specified separately below. The basis study adds 52,698 baseline updates and 9,000 continuation updates. The subsequent causal study adds 364,302 updates: five 24,000-step arithmetic branches, fifteen 5,000-step specificity branches, four original-

Table 20: Stage-level interventions at depth 4, step 53,000, baseline 99.30%. Relocation transports the addition family onto other diagonal frequencies at matched power; phase scrambling randomizes the phase of each conjugate pair. The two stochastic columns show the reported realization (replicate 4 of five); zeroing is deterministic.

Stage	Kind	Relocation	Phase	Zeroed
Block 1 attention write	write	99.30%	99.30%	86.31%
Block 2 attention write	write	99.33%	99.29%	0.94%
Block 3 attention write	write	99.29%	99.30%	0.44%
Block 3 MLP activation	activation	1.66%	0.00%	1.19%
Block 3 MLP write	write	0.95%	0.00%	1.19%
Block 4 attention write	write	3.58%	0.00%	58.44%

Table 21: Direct-readout accuracy by post-block residual, decoding with the unembedding and skipping the remaining blocks. Values are ranges over checkpoints at which the model has at least 95% full-grid accuracy.

Depth	Regime	Earlier blocks	Final block
2	AdamW	0.7–1.1%	100.00%
2	Muon	0.8–5.1%	95.8–100.00%
2	Stable Muon	1.3–5.2%	96.4–99.4%
4	Muon	0.8–18.0%	95.1–99.3%
4	Stable Muon	0.7–18.1%	95.2–98.7%

trajectory extensions totaling 49,302 steps, and four 30,000-step operation or architecture controls. A subsequent horizon extension adds 420,000 updates: each of the five corrected addition branches and the corrected subtraction run continues from step 30,000 to 100,000. The hidden-learning-rate control adds 470,000 updates across five continuations from step 6,000 to 100,000. Verification replays are excluded from these totals. Multiple checkpoints and interventions from one trajectory are dependent observations.

J.1 MONITORING AND MATCHED STATES

The causal branches and fresh baseline replications use a joint event: training and test accuracy must both fall below 90% after sustained grokking. Training accuracy is checked at every update; test accuracy is checked every 100 updates and at every post-grokking state with training accuracy below 90%. Thus every joint event is detected, while brief test-only excursions can fall between evaluations. Six consecutive grid evaluations at or above 95% confirm grokking. The confirmation step is the *last* evaluation of this six-check streak. Original sweep speeds instead report the start of a five-check streak (§2); the criteria therefore differ as well as their reporting points.

Within each seed, matched arithmetic branches restore identical parameters, optimizer buffers, and random-number-generator state. Exact next-update replays reproduce all three. The original prefixes combine historical evaluation grids with densely captured failures; their positive events establish binary incidence. We therefore compare event presence, without comparing their counts, durations, or historical minima against the uniformly monitored branches. The five trajectories were selected for mechanistic study and give no estimate of collapse prevalence over an unrestricted seed population.

The first captured joint failures occur at steps 17,494, 17,721, 18,069, 16,308, and 16,056 for seeds 0–4. Seeds 0 and 4 use exploratory dense replays from steps 16,000 and 15,000, respectively. Seed 0’s earlier 100-step log first detected failure at 17,500; the grid missed seed 4’s brief early failures. Seeds 1–3 were specified together prospectively with the monitoring above and 200 additional updates after capture. The selected failed decoder snapshots for seeds 3 and 4 are later states, 16,309 and 16,060, and retain those identities throughout.

Table 22: Test accuracy (%) at adjacent captured failures. h_0, W_0 precede the update; h_1, W_1 follow it. Each head is evaluated with each feature state, without retraining. Original events are captured under mixed historical monitoring grids (Appendix J.1).

Seed	Before: $W_0 h_0$	Old head: $W_0 h_1$	New head: $W_1 h_0$	After: $W_1 h_1$
0	95.49	96.39	31.86	31.87
1	94.14	94.18	13.56	9.09
2	99.99	100.00	1.01	0.93
3	89.54	91.01	50.80	53.60
4	91.34	98.66	66.75	66.33

J.2 FRESH DECODERS AND THE BASIS HYPOTHESIS

Fresh bias-free linear decoders are fitted to the raw final residual using only the original 3,830 training examples. An uncentered SVD preconditioner retains all 128 dimensions, with singular values floored at 10^{-6} of the largest. Each inner fit estimates its own preconditioner. A 20% split of the training set selects an L_2 penalty on the preconditioned coefficients from $\{0, 10^{-8}, 10^{-6}, 10^{-4}, 10^{-2}\}$ by validation cross-entropy; the selected decoder is then refitted on all training examples. The 8,939 test examples enter only the final evaluation. Fits use float64 logistic regression and L-BFGS-B; convergence and unregularized controls are recorded.

At the selected collapsed checkpoints, decoder accuracies for seeds 0–4 are 98.72%, 98.20%, 100.00%, 98.77%, and 100.00%, against 99.51%, 99.56%, 100.00%, 99.40%, and 100.00% at their references. Probe checkpoints are steps 17,494, 17,721, 18,069, 16,309, and 16,060: the last two occur after the adjacent transitions in Table 22. These results establish substantial surviving linear task information; the small gaps in three seeds leave room for reduced decodability or probe limitations.

For column residuals, an exact global basis account requires one invertible R satisfying $h_1(x) = R h_0(x)$ across the operand grid. With examples stacked as rows, full column rank makes this equivalent to equality of the two feature column spaces. We test train-fitted linear and affine maps, centered reconstruction, rank sensitivity, principal angles, and a separately identified full-domain oracle. Centering by training means prevents a large constant feature component from dominating reconstruction error.

Table 23: Centered held-out reconstruction error (%) for a train-fitted map from later to earlier residuals. Healthy controls match interval length. Reference intervals span 6–508 updates and occur at different optimization stages, so comparisons are within each seed.

Seed	Reference to captured failure	Healthy interval	Adjacent failure update
0	43.63	32.66	1.88
1	43.05	10.82	4.63
2	9.05	2.76	1.10
3	45.73	14.30	10.66
4	25.60	1.18	25.63

Positive controls plant exact residual gauges into the trained network. Eight controls cover condition numbers 1, 3, 10, and 100 in float32 and float64; twenty-four further controls plant observed fitted matrices with their measured orientation and conditioning. Exact compensation and train-fitted compensation recover 100% in every fixed-condition control. For the three fresh adjacent transitions, observed forward-map errors are 4.65%, 1.10%, and 10.77%, against 0.0004–0.0006% for their planted controls. These errors reject exact global equivalence for the measured pairs. Healthy evolution also departs from exact equivalence (Table 23).

Approximate transport remains useful: inverse-linear compensation reaches 95.13%, 100%, and 94.37% at the three fresh first events. Reference timing matters. Seed 0’s head from step 17,200 scores 14.05% on the collapsed residual, while the head from 17,493 scores 96.39%. Its geometric discrepancy from the distant reference is already present before the final failing update. Moreover, the complete real task-only Fourier coefficient matrices have shape 113×128 ; when both have full

row rank, an invertible map between them always exists. For the seed-4 generalization-drop pair at steps 16,000 and 17,000, a fitted task-only map aligns that projection exactly while scoring 4.62% on raw held-out features. This pair is separate from the adjacent failures above.

One-time repairs also have limited persistence. Eighteen continuations resume two captured failures for 500 updates under nine head/state interventions each. Every successful repair retaining optimizer state falls below 90% training accuracy on the next update. Clearing the head’s Adam state improves eventual recovery but permits the immediate relapse. These tests separate instantaneous decodability from stability under continued optimization.

J.3 ARITHMETIC INTERVENTIONS

The accurate loss evaluates $\log(1 + \sum_{j \neq y} \exp(z_j - z_y))$ with `log1p` whenever the correct-class logit is maximal; other examples use ordinary cross-entropy. Model storage, backward matrix arithmetic, and Muon’s update remain float32. This avoids cancellation in the small loss and correct-class derivative, following the numerical concerns of Prieto et al. (2025). All five step-6,000 sources already satisfy the six-check grokking criterion and already contain measurable gradient error. The intervention acts on an existing post-grokking trajectory.

Table 24: Matched arithmetic branches from step 6,000 through 30,000. Original columns show a captured joint failure. Accurate-CE branches have no joint failure; all finish at 100% train and test accuracy. Minima include the starting state.

Seed	Original event step	Original event test (%)	Accurate minimum test (%)
0	17,494	31.872	99.754
1	17,721	9.095	99.843
2	18,069	0.929	98.859
3	16,308	53.597	99.978
4	16,056	66.327	99.843

The specificity study branches each seed at step 15,000 and runs to 20,000 with three corrections. Accurate CE gives 0/5 joint failures; target-derivative repair gives 0/5; per-example zero-sum projection gives 0/5. Every original trajectory has a captured failure in this window. All fifteen corrected branches finish at 100% test accuracy; the minimum measured test accuracy across these branches is 99.978%.

For stock logit gradient g , target repair retains each wrong-class entry and sets $g_y = -\sum_{j \neq y} g_j$, leaving the reported forward loss unchanged. Projection replaces g by $g - \mathbf{1}(\mathbf{1}^\top g)/p$. Both corrections compute their class sums in float64 and cast the result back to float32. Derivative errors are relative L_2 errors against the accurate float64 derivative evaluated on the identical float32 logits. The readout-gradient reference casts this logit derivative to float32 before backpropagating through the shared float32 network. At the fixed seed-4, step-16,000 audit, target repair reduces relative derivative error from 32.02% to 4.19×10^{-7} , whereas projection leaves 31.88%. At step 20,000, projected derivatives retain 52.94–86.07% relative error despite a class-sum residual below 2.1×10^{-8} relative to the reference norm. Thus restoring the class-sum identity can prevent the observed failures over this window even with substantial remaining derivative error. Numerical feature inflation and this projection control build on Liu et al. (2026).

At step 15,000, stock logit-gradient errors are 10.56–21.86% and readout-gradient errors 18.95–43.42%. Passing a class-common gradient error through the same saved Adam state produces a class-centered update difference: its centered norm fraction is 45.85–73.37%, versus $(2.8\text{--}4.1) \times 10^{-16}$ for scalar SGD, at float64 roundoff. These descriptive norm ratios are nonadditive. Accurate-CE branches also have smaller feature means: original mean norms at step 15,000 are 2.22–3.83 times their matched corrected values.

Persistence through the original training horizon. We resume each corrected step-30,000 state through step 100,000, matching the original depth-1 horizon. All model parameters, optimizer buffers, and the PyTorch random state are restored. Twelve direct updates agree bitwise with both the monitored runner and a six-plus-six resumed continuation for all six sources. Each extension

adds 70,000 updates, with training accuracy recorded at every state and test accuracy every 100 steps and whenever training accuracy is below 90%. The final and minimum-accuracy checkpoints reproduce their logged metrics exactly.

Table 25: Corrected-loss runs through step 100,000. Accuracies are percentages. Minima combine the earlier prefix with the extension, counting the shared step-30,000 state once. Addition is corrected from step 6,000; subtraction uses corrected CE from initialization and its minima start at grokking confirmation.

Run	Monitoring start	Min. test	Final test	Joint event
Addition, seed 0	6,000	99.754	100.000	none
Addition, seed 1	6,000	99.843	100.000	none
Addition, seed 2	6,000	98.859	100.000	none
Addition, seed 3	6,000	99.978	100.000	none
Addition, seed 4	6,000	99.843	100.000	none
Subtraction, seed 0	10,500	96.879	99.955	none

All six extensions complete without a joint event. Thus correction suppresses the observed failures through the original training budget. This addresses postponement beyond step 30,000 within that budget; behavior after 100,000 and brief test-only excursions between scheduled evaluations remain outside the measurement. The original stock trajectories already have captured failures by 30,000, which establishes their positive binary incidence at the longer horizon. Their historical monitoring supports no comparison of event counts or durations with these dense branches.

J.4 MEAN AMPLIFICATION AND FINITE-STEP OVERSHOOT

At the checkpoints immediately before the five original failures, feature means over the full operand grid have norms 10,374–12,203. The main-text intervention in Eq. 1 uses the post-update representation’s training mean and adds each component to the rescued logits $W_{U,0}h_1$. Applying the training-estimated mean unchanged to test examples, the mean term alone gives 32.11%, 9.42%, 0.93%, 53.34%, and 66.07% accuracy, close to the full failing updates in Table 24. The centered term alone retains 95.26%, 92.48%, 100%, 87.20%, and 98.46%. Relative reconstruction error is below 3×10^{-16} . This evaluation-only intervention establishes local sufficiency of the product of feature mean and readout displacement. Feature growth and optimizer history remain coupled before that step.

Replacing gradients only at the failing update with accurate gradients preserves all five failures. The state accumulated beforehand matters. For each actual complete parameter displacement $\Delta\theta$, the accurate training gradient satisfies $\nabla L^\top \Delta\theta < 0$, including each group’s contribution. Scaling the displacement by 0.001 or 0.01 lowers training loss in every seed; taking the full step raises it sharply. Both interpolation endpoints reproduce saved parameters exactly. The captured updates locally descend and overshoot at finite scale.

J.5 FEATURE-MEAN TIMELINES IN THE ARCHIVED SEED STUDY

Figure 7 describes the available checkpoints from five original Muon and five selected AdamW trajectories. These archived trajectories are distinct from the fresh, densely captured cohort used for adjacent-update localization. For each checkpoint, we measure $\|\mu\|_2$, where $\mu = p^{-2} \sum_{a,b} h(a, b)$ is the mean of the actual readout-input features over the full operand grid. The archived orthonormal Fourier summaries give

$$\|\mu\|_2 = \frac{\sqrt{P_{\text{total}} f_{\text{DC}}}}{p},$$

since $\hat{h}(0,0) = p\mu$. This measurement uses the unfiltered representation. All 88 plotted rows were independently recomputed; a direct feature-mean check on a saved checkpoint agrees with its orthonormal-DFT calculation to 5.49×10^{-16} relative error.

Muon uses hidden learning rate 0.03 and decay 0.1, embedding AdamW learning rate 0.001 and decay 1, and readout AdamW learning rate 0.00025 and decay 1. The selected AdamW reference

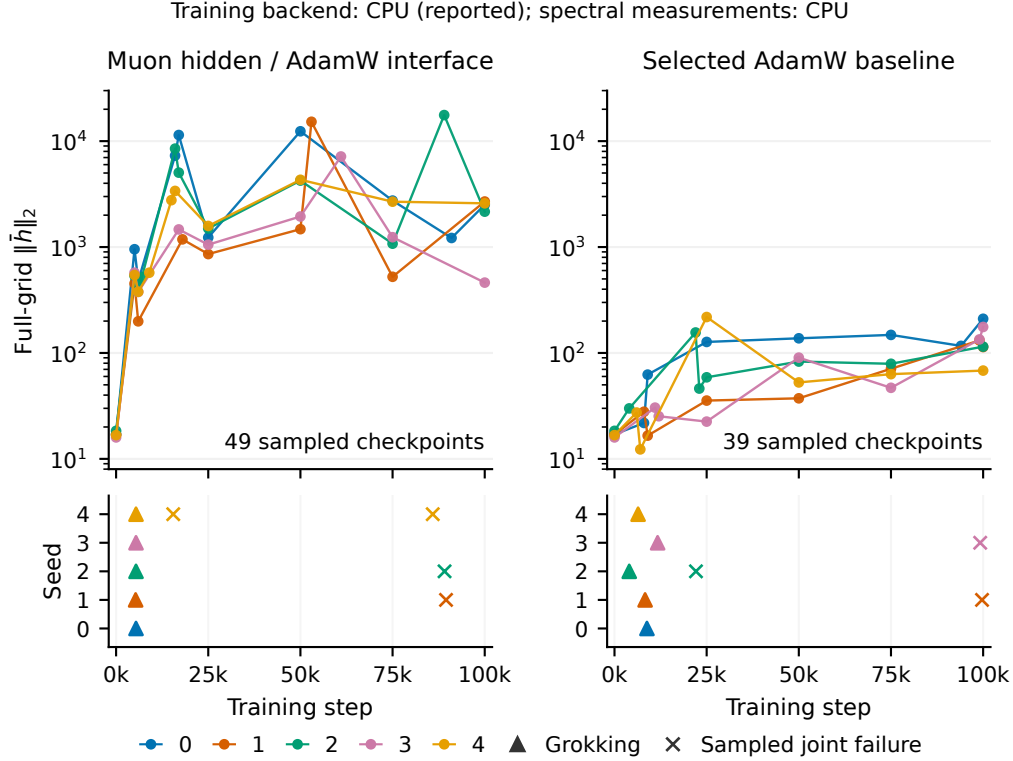


Figure 7: Readout-input feature means in archived Muon and selected AdamW runs. Circles show all available checkpoints: 49 for Muon and 39 for AdamW. Lines connect these irregularly sampled states. Norms use the full operand grid. Lower rasters mark original grokking and joint failures observed on the 100-update grid: four Muon failure observations across three seeds and three AdamW observations across three seeds. This sampling can miss brief events. The plot identifies configurations and the archive’s reported CPU backend; no original runtime manifest is retained. These descriptive trajectories provide context for the matched mean-displacement and arithmetic interventions.

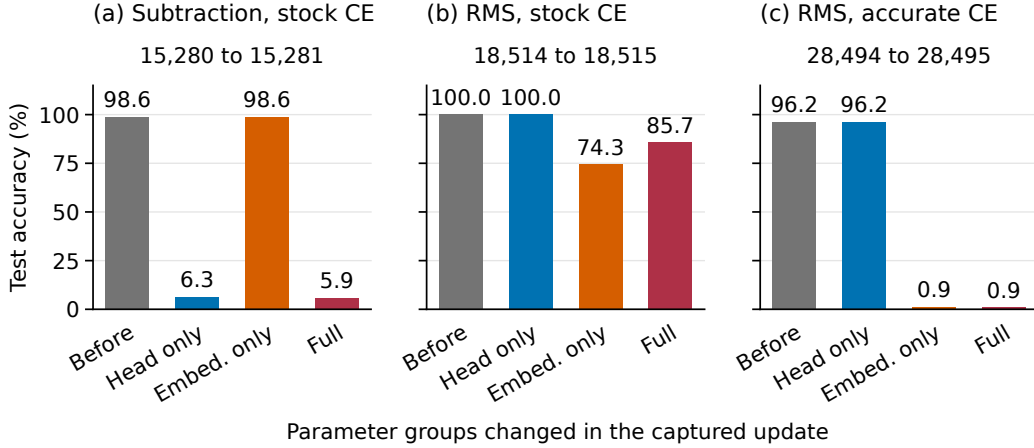


Figure 8: One-step parameter interventions at the first detected joint failure in each operation or architecture control, all at seed 0. Each bar uses the preceding checkpoint with only the indicated parameter group replaced by its following value; “Full” applies all three groups. In subtraction, the head update reproduces most of the damage. In both RMS cases, the embedding update is destructive while the head update preserves accuracy. The accurate-CE RMS failure occurs with an accurate loss derivative. These are first-event measurements at the displayed steps; the later post-grokking minima are reported separately in Table 26.

uses learning rate 0.001 and decay 3 throughout. The original archive reports CPU training, and its Fourier analysis runs on CPU. Checkpoint selection includes phase landmarks and low-accuracy states; intermediate feature means are unmeasured. The larger recorded Muon means are descriptive evidence about these configurations and sampled trajectories. The controlled intervention in §4.4 addresses the local effect of multiplying the observed readout displacement by the feature mean.

J.6 SUBTRACTION AND NORMALIZED ARCHITECTURE

Four additional runs use seed 0 and a fixed 30,000-step horizon: a stock/accurate-CE pair on subtraction and a pair on addition with gain-free RMS normalization before attention, before the MLP, and before the readout ($\epsilon = 10^{-6}$). Parameter initialization and optimizer settings are matched, with no added parameters or hyperparameter search.

Table 26: Operation and architecture controls. Confirmation is the end of the six-check grokking streak. First-event test accuracies and post-grokking minima are separate measurements; both RMS runs subsequently recover to 100%. Each row is one seed.

Condition	CE	Confirmation	First event	Event test (%)	Min. test (%)
Subtraction	Stock	9,300	15,281	5.851	5.851
Subtraction	Accurate	10,500	none	n/a	96.879
RMS addition	Stock	15,700	18,515	85.658	0.481
RMS addition	Accurate	19,900	28,495	0.884	0.526

Subtraction reproduces acute readout sensitivity: updated features with the preceding head score 98.81%, the mean term alone 5.84%, and the centered term 98.76%. At step 30,000, accurate CE has no joint event and test accuracy is 99.966%. Its continuation through 100,000 is reported in Table 25.

In the RMS runs, the embedding update causes the acute loss of accuracy. In the accurate-CE event, test accuracy falls from 96.208% to 0.884%; its logit derivative has relative error 9.48×10^{-8} and no vanished correct-class entries. Updating only the head retains 96.208%, updating only embeddings gives 0.940%, and freezing the head still gives 0.895%. The minimum reaches 0.526% at step

28,503; the stock-RMS minimum is 0.481% at 28,540. Exact update replay passes. Both RMS event directions locally descend and lower loss at scales 0.001 and 0.01, then overshoot at full scale. This historical counterexample bounds the readout/feature-mean account to the measured unnormalized cases and rules out universal prevention by accurate loss arithmetic. Appendix J.9 adds four prospective accurate-CE RMS seeds and applies the same event analysis to every captured failure.

J.7 POST-GROKING HIDDEN-LEARNING-RATE REDUCTION

We restore the same five original step-6,000 checkpoints used for the arithmetic branches, including the full model, optimizer buffers, and random-number-generator state. Immediately after restoration, we reduce the hidden Muon learning rate from 0.03 to 0.003 and hold it fixed through step 100,000. Embeddings and readout remain trainable at learning rates 0.001 and 0.00025, respectively. All branches retain stock cross-entropy and the original weight-decay coefficients: 0.1 for the hidden matrices and 1 for embeddings and readout. Each continuation adds 94,000 updates, for 470,000 in total.

The primary outcome is whether a seed has any state with both training and held-out test accuracy below 90%. Training accuracy is recorded at every update, including the source and final states. Test accuracy is evaluated every 100 updates and whenever training accuracy is below 90%, matching the corrected-loss protocol. This gives 94,001 recorded training states per seed. Test-only excursions between scheduled evaluations can be missed.

Table 27: Hidden-LR continuations through step 100,000. Every seed has a joint failure. Test accuracies are percentages, and minima include the starting state. Below-95% counts include all test evaluations under the stated monitoring rule; the scheduled subset uses only the 100-step grid. These counts describe the new branches.

Seed	First joint failure step	Minimum sampled test	Evaluations below 95%	Scheduled subset	Final test
0	18,767	0.8614	105	13	100.0000
1	18,476	4.7209	88	20	100.0000
2	19,439	0.8950	135	14	99.6644
3	17,041	4.8999	52	22	99.8881
4	17,578	0.8166	103	13	100.0000

All five reduced-LR branches experience a joint failure, compared with 5/5 original stock trajectories and 0/5 corrected-loss branches through the same horizon. We compare binary seed-level incidence because the original trajectories have different historical monitoring. The reduced-LR branches first fail between steps 17,041 and 19,439; seeds 0, 2, and 4 reach approximately chance-level test accuracy ($1/113 \approx 0.885\%$). All recover to at least 99.6644% final test accuracy. Their recovery illustrates why final accuracy alone misses these failures.

Table 28: Sampled diagnostics for the hidden-LR branches. Each seed has 95 measurements at 1,000-step intervals, including steps 6,000 and 100,000. Feature means cover the full operand grid. CE error is the relative L_2 error of the stock float32 training-logit derivative against the accurate float64 derivative at the same logits, expressed as a percentage. Maxima are over the sampled diagnostic states.

Seed	Feature-mean norm			CE derivative error (%)		
	Start	Final	Maximum	Start	Final	Maximum
0	310.8	292.6	4,282.5	1.440	0.030	38.558
1	459.5	505.3	8,783.8	1.500	0.180	46.273
2	355.1	1,096.5	8,100.4	0.967	22.837	36.585
3	411.9	955.8	8,878.5	2.228	5.408	48.431
4	434.1	445.1	3,596.0	1.748	0.027	63.645

Every reduced-LR branch exhibits substantial feature-mean growth and loss-derivative error at sampled states (Table 28). The accompanying artifacts retain training-set feature means, raw trajectories, checkpoint replays, and the detailed audits. We have not localized the parameter group responsible for these reduced-LR failures, so the evidence establishes joint failure without attributing it to the acute readout mechanism identified in the original branches. Ordinary learning-rate reduction also scales the decoupled weight-decay step: the hidden per-update decay factor changes from $1 - 0.03 \times 0.1 = 0.997$ to $1 - 0.003 \times 0.1 = 0.9997$. The tested tenfold reduction is therefore a stabilization control. It is insufficient to provide the stability achieved by corrected loss arithmetic for these five matched continuations through step 100,000.

J.8 INITIALIZATION AND MEMORIZATION DECODER CONTROLS

We apply the same decoder to initialization and one memorization-phase state of each of the five unnormalized trajectories in Appendix J.2. Checkpoint selection uses the recorded native trajectory: the first of five consecutive scheduled 100-step training evaluations at or above 99.9%, requiring test accuracy below 95% at that state and a step before original grokking. For seeds 0–4, this selects steps 200, 400, 200, 100, and 200; the corresponding fifth training checks occur at 600, 800, 600, 500, and 600. The selection rule, checkpoint identities, and validation indices were fixed before decoder fitting. The initial registration and its pre-fitting amendment to the paper’s sustained-memorization criterion are retained.

Replaying each saved initialization through step 1,000 reproduces the original model, optimizer, and CPU RNG states exactly, as well as all ten scheduled evaluation rows. These replays recover the selected early checkpoints without substituting a new trajectory. Their original five-check grokking starts are 5,500, 5,500, 5,500, 5,300, and 5,400, respectively; the historical speed sweep has its own run identities.

Features are captured by a pre-hook on the readout, with no centering or Fourier filtering. All four checkpoints within each seed use the same internal validation indices. The bias-free multinomial logistic regression starts from zero and fits in float64. Every candidate estimates its uncentered, full-dimensional SVD transform using only its fitting subset, with a relative singular-value floor of 10^{-6} . Validation cross-entropy selects the penalty from $\{0, 10^{-8}, 10^{-6}, 10^{-4}, 10^{-2}\}$; exact ties select the smaller penalty. The selected model is refitted from zero on all original training examples with a newly fitted transform. L-BFGS-B uses 500 candidate iterations and 1,000 refit iterations, $\text{ftol} = 10^{-10}$, $\text{gtol} = 10^{-7}$, at most three times the iteration budget in function evaluations, 20 line-search steps, and history size 10. These settings match the existing decoder procedure.

The selected decoder reaches 0.04–0.23% held-out accuracy at initialization and 0.02–1.12% at memorization, compared with 99.40–100% at the existing references and 98.20–100% at the failed states (Table 29). All 20 selected refits converge. Unregularized memorization refits also converge; all five initialization refits reach the 1,000-iteration limit, with 0.18–0.62% held-out accuracy. Candidate objectives, gradient norms, iteration counts, and termination messages accompany the saved coefficients and transforms. Independently recomputing training and test predictions from these arrays reproduces all 40 new accuracy measurements.

Table 29: Training-only decoder controls on the five trajectories used for the reference and failed-state probes. All accuracies are percentages. Memorization is the first of five consecutive scheduled training evaluations at $\geq 99.9\%$, before grokking and with test accuracy below 95%. Refit reports convergence and iterations; Cand. gives converged candidate fits out of five. The ten reference and failure results are unchanged.

Seed	State	Step	Native train	Native test	Decoder test	Refit	Cand.
0	Initialization	0	0.76	0.84	0.12	Yes (10)	3/5
0	Memorization	200	99.95	0.08	0.02	Yes (36)	5/5
0	Reference	17,200	100.00	99.12	99.51	Yes (19)	5/5
0	Failed	17,494	48.25	31.87	98.72	Yes (18)	5/5
1	Initialization	0	0.81	0.92	0.12	Yes (10)	2/5
1	Memorization	400	99.97	0.22	0.25	Yes (33)	5/5
1	Reference	17,300	100.00	99.19	99.56	Yes (19)	5/5
1	Failed	17,721	15.67	9.09	98.20	Yes (17)	5/5
2	Initialization	0	0.73	0.89	0.07	Yes (10)	3/5
2	Memorization	200	99.95	0.49	0.55	Yes (34)	5/5
2	Reference	18,000	100.00	100.00	100.00	Yes (19)	5/5
2	Failed	18,069	0.99	0.93	100.00	Yes (18)	5/5
3	Initialization	0	0.86	0.91	0.23	Yes (11)	3/5
3	Memorization	100	100.00	0.09	0.07	Yes (37)	5/5
3	Reference	15,800	100.00	99.07	99.40	Yes (19)	5/5
3	Failed	16,309	42.64	32.12	98.77	Yes (18)	5/5
4	Initialization	0	1.04	0.83	0.04	Yes (10)	2/5
4	Memorization	200	99.95	1.02	1.12	Yes (33)	5/5
4	Reference	16,050	100.00	100.00	100.00	Yes (19)	5/5
4	Failed	16,060	21.78	18.13	100.00	Yes (17)	5/5

The early controls support the interpretation that the failed checkpoints retain task information made accessible by training. They leave its precise emergence between checkpoints unresolved. Low early accuracy under this bounded decoder procedure does not establish mathematical absence of task information, and high failed-state accuracy alone does not identify a complete arithmetic algorithm or exact preservation of the earlier representation.

J.9 ACCURATE-CE RMS REPLICATION AND EVENT LOCALIZATION

Registered cohort and execution. The historical accurate-CE RMS run in Appendix J.6 is a seed-0 CPU pilot with a 30,000-step horizon. We extend it with prospectively registered seeds 1–4 on CPU, preserving the pilot’s backend and PyTorch 2.13.0 implementation. All runs use addition modulo 113, the original seeded 30% split, one block, width 128, four heads, MLP width 512, ReLU, and bias-free projections. Gain-free RMS precedes attention, the MLP, and the readout, with $\epsilon = 10^{-6}$. The model and all five Newton–Schulz iterations remain float32. Hidden Muon uses learning rate 0.03, momentum 0.95, Nesterov updates, and decay 0.1. Embedding and readout AdamW use learning rates 0.001 and 0.00025, respectively, decay 1, betas (0.9, 0.999), and source $\epsilon = 10^{-8}$. Source values for the remaining optimizer options are retained. The cancellation-resistant cross-entropy is active from initialization.

The execution manifest records Python 3.14.6, NumPy 2.5.2, SciPy 1.18.0, macOS ARM64 with Accelerate BLAS, one PyTorch intra-op and inter-op thread, deterministic algorithms, and highest float32 matrix-multiplication precision. Initial and event replays reproduce the historical seed-0 model exactly under this environment. Hashed source snapshots, the complete optimizer configuration, and the registered protocol accompany the study.

Monitoring and capture. Training accuracy is inspected at every state, using exact correct/total counts. Testing occurs every 100 updates and at every post-confirmation state with training accuracy below 90%. Six consecutive scheduled test evaluations at or above 95% confirm grokking on the sixth check. Each prospective run stops at its first subsequent joint train/test failure below 90%, or at step 100,000. Outcomes distinguish confirmed grokking with failure, confirmed grokking

with event-free completion, failure to confirm grokking, and technical interruption. Minima and evaluation counts over the post-confirmation window include the confirmation evaluation itself.

Checkpoints preserve initialization, confirmation, every 1,000 updates, and the terminal state. After confirmation, the immediately preceding model, optimizer, and RNG states are retained for event capture. Prospective checkpoints include PyTorch CPU, Python, and NumPy RNG states; GPU RNG is unused and explicitly recorded as such. The historical checkpoint format preserves PyTorch RNG only. Complete-update replay verifies bitwise model and optimizer equality, the available RNG states, and the resulting event measurements. Any interruption requires exact saved-state recovery with documented replay overlap.

Identical event analysis. For every captured transition, we compare the accurate float32 loss derivative with an analytic float64 reference on the identical stored float32 training logits. Saved arrays support absolute L_2 and maximum error, relative L_2 error, reference norm, and class-sum residuals. All eight old/new combinations of embeddings, hidden matrices, and readout are evaluated on the original training and test splits, with exact accuracy counts and accurate CE. Fresh decoders are fitted immediately before and after the update, using the actual input to the unembedding after final RMS normalization. The fitting, validation, and convergence protocol is identical to Appendix J.8, with validation indices shared within each seed and unregularized sensitivity fits retained.

Historical pilot reanalysis. At steps 28,494 $\rightarrow$ 28,495, held-out accuracy changes from 8,600/8,939 = 96.21% to 79/8,939 = 0.88%. All four swaps retaining the preceding embeddings score 96.21–96.41%; all four using the updated embeddings score 0.88–0.94%. The pre-update derivative’s relative L_2 error is 9.48×10^{-8} , with absolute error 4.06×10^{-10} against reference norm 4.28×10^{-3} . Its class-sum residual has L_2 norm 4.30×10^{-10} . Exact complete-update replay passes.

The training-only decoder gives 97.37% before this update and 1.58% afterward; both validation-selected refits converge. The unregularized control gives 95.83% before and 2.62% afterward, with the latter reaching the 1,000-iteration limit. Thus this pilot does not support the high failed-state linear accessibility found in the unnormalized cohort. The bounded fits leave open whether another linear fitting procedure could recover more information. The historical run’s later recovery and minimum remain separately reported in Table 26.

Prospective outcomes. All four new seeds confirm grokking and reach a joint failure, at steps 15,095, 15,789, 14,271, and 15,136 for seeds 1–4, respectively (Table 30). These runs add 60,291 training updates. No run finishes event-free, fails to confirm grokking, or has a technical interruption. Every terminal checkpoint reproduces its recorded metrics, and every complete-update event replay passes the model, optimizer, and RNG checks.

Table 30: Accurate-CE RMS outcomes. Seed 0 is the historical CPU pilot; seeds 1–4 form the prospective CPU cohort. Prospective endpoints are first joint failure or step 100,000. The pilot continued to 30,000; its displayed endpoint is the separately analyzed first event.

Seed	Confirmation	Event / endpoint	Outcome
0 (pilot)	19,900	28,495	Joint failure
1	11,500	15,095	Joint failure
2	12,600	15,789	Joint failure
3	10,000	14,271	Joint failure
4	12,600	15,136	Joint failure

At each prospective event, replacing only embeddings is sufficient for joint failure. With old embeddings, every combination of hidden matrices and readout retains at least 97.87% test accuracy. The hidden update partly offsets the embedding damage in seeds 2 and 3: embeddings-only accuracy is 53.73% and 57.93%, while applying both embedding and hidden updates gives 54.55% and 62.86%. Readout replacement changes test accuracy by at most 0.06 percentage points across these paired combinations. The embedding update therefore supplies the acute damage at all four captured transitions, with smaller interactions from the other groups.

All selected decoder refits before and after the five analyzed events converge. In the prospective cohort, decoder accuracy falls from 98.15–100% before the update to 2.65–93.82% afterward. Seed

Table 31: Adjacent RMS events and training-only decoders. Accuracies are percentages; before means one update before the displayed event. Decoder inputs include final RMS normalization. The last column gives selected-refit convergence before/after. Seed 0 is the historical pilot; seeds 1–4 are prospective.

Seed	Event step	Native test		Decoder test		Converged
		Before	After	Before	After	
0	28,495	96.21	0.88	97.37	1.58	Yes / Yes
1	15,095	99.70	1.63	100.00	17.35	Yes / Yes
2	15,789	100.00	54.55	100.00	64.86	Yes / Yes
3	14,271	100.00	62.86	100.00	93.82	Yes / Yes
4	15,136	97.87	1.03	98.15	2.65	Yes / Yes

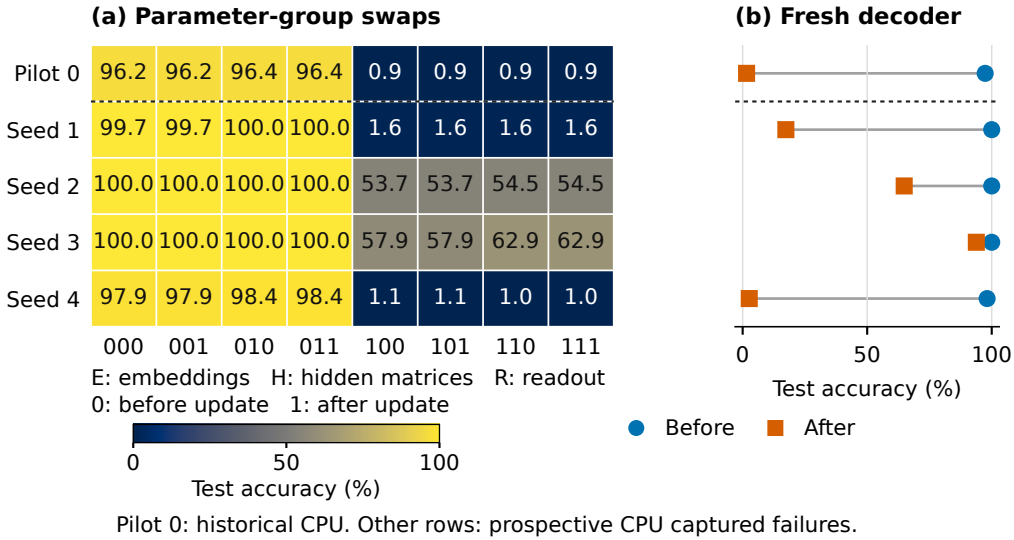


Figure 9: Accurate-CE RMS failures, with the historical pilot separated from four prospective seeds. **Left:** held-out accuracy for all eight old/new parameter-group combinations; mask order is embeddings, hidden matrices, readout. 0 denotes the preceding state and 1 the following state. Every updated-embedding combination gives a joint failure, while old-embedding combinations retain high accuracy. **Right:** fresh training-only decoder accuracy immediately before and after each update, using features after final RMS. All selected refits converge. Exact counts, losses, and sensitivity fits are recorded separately.

3 retains substantial accessibility, whereas seeds 1 and 4 have much lower recovery. Thus the RMS events establish an architectural boundary for both the readout locus and the uniformly high failed-state decoder recovery in the unnormalized cohort. They leave the upstream cause of the embedding displacement unresolved. Tables 35 and 34 retain convergence and derivative diagnostics; full solver records and arrays accompany the paper.

Each prospective run has one sampled post-confirmation test evaluation below 95%, its terminal joint failure. Including confirmation, the four runs have 37, 33, 44, and 27 post-confirmation test evaluations. Their minimum sampled test accuracies equal their terminal accuracies: 1.63%, 54.55%, 62.86%, and 1.03%. These event-stopped summaries have different horizons from the pilot's subsequent recovery and from the 100,000-step unnormalized continuations.

Table 32: All eight parameter-group swaps at captured RMS failures. Seed 0 is the historical pilot; seeds 1–4 are prospective. Mask order is embeddings, hidden matrices, readout; 0 retains the preceding state and 1 uses the following state. Accuracy is in percent. CE is accurate float64 evaluation of the resulting float32 logits. Raw records additionally retain exact counts and float32 accurate CE.

Seed	EHR	Train	Test	Train CE	Test CE
0	000	95.87	96.21	0.34207	0.32198
0	001	95.87	96.21	0.33814	0.32056
0	010	96.50	96.41	0.13595	0.20983
0	011	96.55	96.41	0.13393	0.2092
0	100	1.15	0.94	6.7289	6.7765
0	101	1.12	0.93	6.7313	6.7788
0	110	1.20	0.89	6.808	6.8589
0	111	1.17	0.88	6.811	6.8617
1	000	99.53	99.70	0.0091396	0.0054243
1	001	99.53	99.71	0.008633	0.0051797
1	010	100.00	100.00	2.9798e-05	3.6247e-05
1	011	100.00	100.00	2.945e-05	3.5891e-05
1	100	1.10	1.61	5.8687	5.8818
1	101	1.10	1.61	5.8695	5.8827
1	110	1.07	1.63	5.8567	5.8698
1	111	1.07	1.63	5.8575	5.8707
2	000	100.00	100.00	6.4328e-05	5.5268e-05
2	001	100.00	100.00	6.324e-05	5.477e-05
2	010	100.00	100.00	2.8588e-05	3.2462e-05
2	011	100.00	100.00	2.857e-05	3.2467e-05
2	100	54.57	53.73	3.1082	3.105
2	101	54.57	53.73	3.1082	3.105
2	110	55.22	54.55	3.0872	3.0822
2	111	55.22	54.55	3.0872	3.0823

Table 33: All eight parameter-group swaps at captured RMS failures. Seed 0 is the historical pilot; seeds 1–4 are prospective. Mask order is embeddings, hidden matrices, readout; 0 retains the preceding state and 1 uses the following state. Accuracy is in percent. CE is accurate float64 evaluation of the resulting float32 logits. Raw records additionally retain exact counts and float32 accurate CE.

Seed	EHR	Train	Test	Train CE	Test CE
3	000	100.00	100.00	0.00091077	0.0010679
3	001	100.00	100.00	0.00090878	0.0010657
3	010	100.00	100.00	0.00026494	0.00030755
3	011	100.00	100.00	0.00026434	0.00030686
3	100	58.96	57.93	2.0143	2.1057
3	101	58.96	57.93	2.014	2.1054
3	110	64.15	62.86	1.8055	1.8808
3	111	64.15	62.86	1.8052	1.8805
4	000	97.65	97.87	0.14149	0.12772
4	001	97.65	97.87	0.13947	0.12727
4	010	99.37	98.36	0.04257	0.083994
4	011	99.40	98.41	0.041476	0.083846
4	100	0.84	1.05	6.3598	6.3671
4	101	0.84	1.05	6.3556	6.3626
4	110	0.89	1.03	6.3563	6.3669
4	111	0.89	1.03	6.3517	6.3621

Table 34: Accurate-CE derivative diagnostics on identical stored float32 training logits. Seed 0 is the historical pilot; seeds 1–4 are prospective. Errors compare the full mean-loss derivative with the analytic float64 reference. Class-sum residual is the L_2 norm across examples of the sum over classes. Maximum absolute errors and reference class-sum residuals are retained in the supporting records.

Seed	State	Relative error	Absolute error	Reference norm	Class-sum residual
0	Before	9.48×10^{-8}	4.06×10^{-10}	4.28×10^{-3}	4.30×10^{-10}
0	After	5.08×10^{-8}	8.83×10^{-10}	1.74×10^{-2}	1.55×10^{-9}
1	Before	8.12×10^{-8}	1.00×10^{-10}	1.24×10^{-3}	1.18×10^{-10}
1	After	3.88×10^{-8}	6.35×10^{-10}	1.63×10^{-2}	1.45×10^{-9}
2	Before	4.76×10^{-8}	1.21×10^{-12}	2.54×10^{-5}	3.67×10^{-13}
2	After	4.86×10^{-8}	6.02×10^{-10}	1.24×10^{-2}	9.58×10^{-10}
3	Before	8.99×10^{-8}	1.92×10^{-12}	2.13×10^{-5}	9.97×10^{-13}
3	After	5.04×10^{-8}	5.98×10^{-10}	1.19×10^{-2}	8.80×10^{-10}
4	Before	5.28×10^{-8}	1.40×10^{-10}	2.64×10^{-3}	2.36×10^{-10}
4	After	4.26×10^{-8}	7.16×10^{-10}	1.68×10^{-2}	1.37×10^{-9}

Table 35: RMS decoder convergence and unregularized sensitivity. Seed 0 is the historical pilot; seeds 1–4 are prospective. Candidate convergence reports successful validation-candidate fits out of five; selected-refit convergence is in Table 31. Unregularized accuracy is held-out percent. Complete termination messages, objectives, iteration counts, and gradient norms accompany every fit.

Seed	State	Selected λ	Candidates	Unreg. test	Unreg. converged
0	Before	1.00×10^{-4}	5/5	95.83	Yes
0	After	1.00×10^{-2}	3/5	2.62	No
1	Before	1.00×10^{-8}	5/5	100.00	Yes
1	After	1.00×10^{-2}	5/5	11.05	Yes
2	Before	0	5/5	100.00	Yes
2	After	1.00×10^{-2}	5/5	55.35	Yes
3	Before	0	5/5	100.00	Yes
3	After	1.00×10^{-6}	5/5	88.66	Yes
4	Before	1.00×10^{-4}	5/5	96.21	Yes
4	After	1.00×10^{-2}	3/5	4.13	No

K TASK STRUCTURE IN FOURIER FILTERING

K.1 FULL-FAMILY PROJECTION IS AN ORBIT AVERAGE

Write a Fourier mode as $\chi_{k,l}(a, b) = \exp(2\pi i(ka + lb)/p)$. Averaging its translations gives

$$\frac{1}{p} \sum_{t=0}^{p-1} \chi_{k,l}(a+t, b-t) = \chi_{k,l}(a, b) \frac{1}{p} \sum_{t=0}^{p-1} \exp\left(\frac{2\pi i(k-l)t}{p}\right) = \mathbf{1}_{k=l} \chi_{k,l}(a, b). \quad (3)$$

Linearity proves Eq. 2 for each feature channel. The surviving modes include $(0, 0)$. For subtraction, averaging $(a+t, b+t)$ instead retains $k+l=0$ modulo p . With a linear head, $W_U Ph = P(W_U h)$; an affine head also commutes with this average because averaging preserves its constant bias. Excluding the constant mode gives the orbit average minus the global mean.

The full-family filter therefore supplies the equivalence relation defining the answer. It can pool logits from training and held-out examples with that answer. This is part of the intervention itself, even though no labels are passed to the FFT routine. Its accuracy measures an answer-conditioned average and requires separate evidence to identify the algorithm implemented by the unfiltered model.

K.2 A LOOKUP-TABLE MEMORIZER

We use the repository’s five seeded splits, each with 3,830 training and 8,939 held-out examples at $p = 113$. Let T denote the training set and e_s the one-hot vector for class s . The predictor stores $z(a, b) = e_{a+b}$ on T and outputs zero on every unseen pair. If n_s is the number of training examples with sum s ,

$$(P_{\text{add}} z)(a, b) = \frac{n_{a+b}}{p} e_{a+b}. \quad (4)$$

Every answer orbit is covered in each split, so filtering yields 100% accuracy. For a uniform split of $m = 3,830$ out of $N = p^2$ pairs, an orbit is uncovered with probability $q = \binom{N-p}{m} / \binom{N}{m}$. A union bound gives $\Pr(\text{some orbit uncovered}) \leq pq < 2.89 \times 10^{-16}$. Before filtering, all held-out logits are tied. We use the smallest class index as the deterministic tie rule, matching the implementation’s argmax; uniform random tie breaking has expected accuracy $1/113 = 0.885\%$.

Table 36: Addition memorizer on the original seeded splits. Correct counts use the 8,939 held-out examples. The filtered logits have a unique correct maximum on every example.

Seed	Raw correct	Raw accuracy (%)	Filtered correct	Filtered accuracy (%)
0	79	0.884	8,939	100.000
1	68	0.761	8,939	100.000
2	79	0.884	8,939	100.000
3	81	0.906	8,939	100.000
4	77	0.861	8,939	100.000

The same control on subtraction reaches 100% after the corresponding filter in all five seeds. Omitting the constant mode also retains perfect accuracy in these controls. A seeded bijection of output labels preserves the perfect filtered result when evaluated against those relabeled classes, because the projection still pools within answer orbits. Averaging training donors alone is sufficient; averaging only held-out donors gives zero logits for the memorizer.

These controls establish that full-family filtered accuracy can be perfect without a learned generalizing arithmetic algorithm. The 95.14% filtered accuracy at AdamW step 3,000 therefore requires independent evidence to establish when an internal arithmetic algorithm was learned. Evidence for surviving information at the collapsed checkpoints comes from the raw-feature decoder tests and adjacent-state swaps in §4.2.

K.3 SPARSE SUFFICIENCY, ABLATION, AND PERTURBATION CONTROLS

A second lookup stores $(p/n_c)e_c$ for a training example with label c , using training-label counts only, and zero for unseen pairs. Its full projection is exactly e_{a+b} . Retaining any one nonzero diagonal

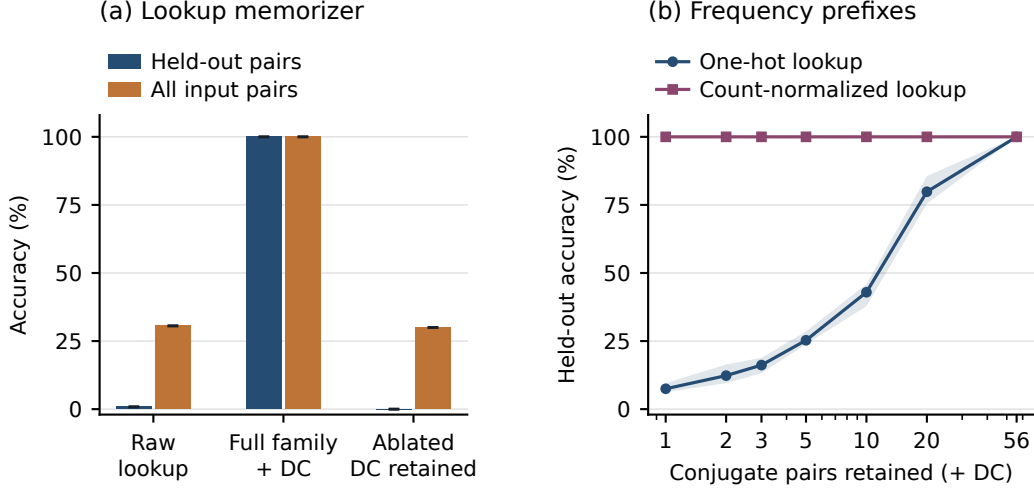


Figure 10: Task structure supplied by Fourier filtering at $p = 113$. **(a)** The lookup memorizes the 3,830 training answers and returns zero on the 8,939 held-out inputs. Full-family filtering with DC gives 100% accuracy. Family ablation retains DC and yields 0% held-out accuracy and 29.99% full-grid accuracy. Bars show means over five splits; whiskers span their range. **(b)** Training-count normalization allows any one diagonal conjugate pair plus DC to classify every input correctly. The ordinary lookup’s pair powers are mathematically tied; its prefixes use increasing frequency as a fixed tie break. Lines show means and shading spans the seed range.

conjugate pair and DC gives class- c logits

$$z_k(s, c) = \frac{1 + 2 \cos(2\pi k(s - c)/p)}{p}, \quad s = (a + b) \bmod p. \quad (5)$$

For prime $p = 113$ and nonzero k , the unique maximum is at $c = s$. All 56 individual pairs give 100% training, held-out, and full-grid accuracy on all five splits (280 checks against the closed form). Thus sparse filtered sufficiency also admits a lookup-only predictor. The construction can use a 113-dimensional lookup representation with an identity head, padded to the transformer’s 128-dimensional residual width.

The ordinary one-hot lookup has equal mathematical power in all diagonal conjugate pairs. With a fixed ascending-frequency tie break, five-pair filtering gives 23.82–28.57% held-out accuracy; sorting the numerically computed powers instead gives 63.69–87.46%. These ranges expose the sensitivity of the selected frequencies to floating-point tie breaking in this control. Figure 10 uses the fixed ordering. The count-normalized lookup reaches 100% at every tested prefix size 1, 2, 3, 5, 10, 20, 56.

The original family-ablation convention is $z - Pz + \bar{z}$, retaining DC. For both lookups, it preserves all training predictions and gives 0% held-out accuracy, hence $3,830/12,769 = 29.9945\%$ full-grid accuracy. The trained models’ near-chance full-grid ablations in Table 4 contain additional information absent from this simple control. Full-family sufficiency and low held-out ablation accuracy can nevertheless occur together under memorization.

We also apply 20 fixed-seed replicates per split and lookup for pair-global phases, channelwise phases, and derangements of the 56 pair coefficients, preserving DC, conjugate symmetry, and total nonconstant family power. For the ordinary lookup, mean held-out accuracies of the isolated family are 1.0755%, 0.9426%, and 3.0680%, respectively; for the count-normalized lookup they are 0%, 0.8850%, and 3.1555%. Each mean pools 20 perturbations per split across five splits; raw replicate distributions are released. The sensitivity observed in the trained models remains a functional measurement of their projected components. These controls show why that sensitivity alone leaves the internal algorithm unidentified.

K.4 RESTRICTING THE AVERAGING DONORS AT CAPTURED FAILURES

Using both endpoints of the five adjacent failure pairs in Table 22, we recompute logits from the saved raw residuals and native heads, multiplying in float64. We average only held-out inputs with the same answer, excluding the evaluated input itself. This removes both training donors and self contribution. Input order, labels, and masks are checked against the original seeded splits.

Table 37: Held-out accuracy (%) before and after each captured update. “Average” uses held-out donors with the same answer and excludes the evaluated example. The operation still supplies the grouping.

Seed	Before		After	
	Raw	Average	Raw	Average
0	95.49	100.00	31.87	30.46
1	94.14	100.00	9.09	0.93
2	99.99	100.00	0.93	0.88
3	89.54	99.12	53.60	65.29
4	91.34	98.18	66.33	71.25

Before these updates, high accuracy survives removal of training donors, establishing task-aligned signal in the logits of unseen examples beyond the ordinary lookup control. The donor-restricted operation still uses the answer orbits and remains a task-informed diagnostic. These checkpoints are distinct from the early AdamW checkpoint at step 3,000; this analysis makes no claim about its donor dependence.

Independent FFT, translated-grid sums, and orbit reductions agree in float64 at moduli 7 and 113. We also check projector idempotence, constant-mode conventions, readout commutation, coverage of every answer orbit, and the single-pair closed form. All scripts, protocols, integer counts, perturbation outcomes, and input hashes accompany the release.

L PROJECTION INTERFERENCE AND SPECTRAL ANALYSES

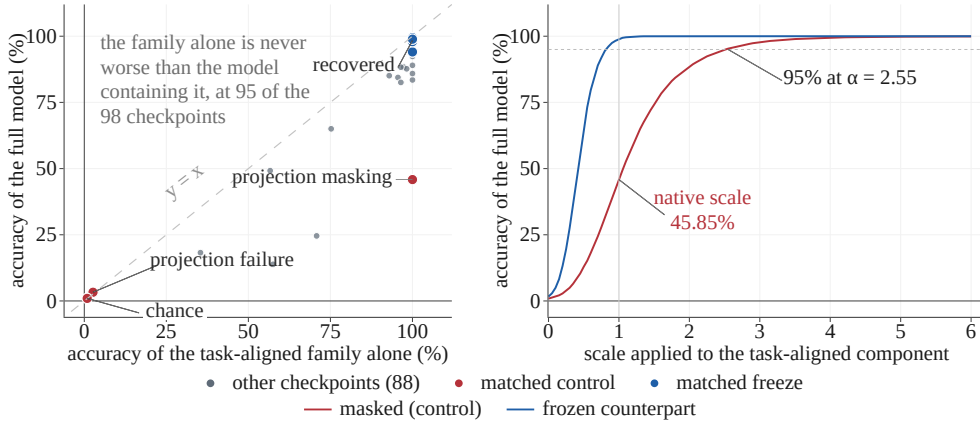


Figure 11: **Left:** task-aligned family accuracy (including the retained constant mode) against full-model accuracy over 98 checkpoints from the original studies, both evaluated through the current readout. *Projection failure* denotes low filtered accuracy; *projection masking* sits at the lower right, where the task-informed projection is accurate and the full model fails. At 95 of 98 checkpoints the family alone is at least as accurate as the full model, and the three exceptions exceed it by under a point. **Right:** rescaling the non-constant task-aligned component of the masked checkpoint by α , with the rest of the representation and the readout untouched.

Filtering the final residual gives an answer-conditioned average through the current readout (Eq. 2). Comparing it with the full model measures interference from the remainder. In *projection failure*, filtering gives 2.65% and then chance, and the family has fallen out of the top twenty power pairs almost entirely. Fresh-decoder recovery in §4.2 provides a separate test of surviving task information. In *projection masking*, the filtered representation achieves 100% accuracy over the full operand grid. The full model reaches 45.85%. The projection averages to a correct answer for every orbit; the full prediction also includes the remainder. One matched trajectory passes through failure, full recovery, and masking; its frozen counterpart retains high filtered accuracy and substantial family power (Appendix H).

Table 38: Correct-class margin at the masked checkpoint, decomposed by linearity of the bias-free readout. The competitor is the highest-scoring incorrect class under the full model, fixed across components. The constant mode belongs to the remainder. The final column is the fraction of examples with a positive component margin.

Branch	Component	Mean	On correct	On wrong	Fraction > 0
Control	full	-0.71	2.15	-3.13	0.4585
Control	task-aligned	+5.72	5.30	6.08	1.0000
Control	remainder	-6.43	-3.15	-9.21	0.0475
Freeze	full	+11.61	11.76	-1.67	0.9889
Freeze	task-aligned	+19.28	19.30	17.18	1.0000
Freeze	remainder	-7.67	-7.54	-18.86	0.0622

Masking is a competition on amplitude. Because the unembedding is linear and bias-free, the correct-class margin decomposes exactly into the task-aligned contribution and the remainder (Table 38). Against the full model’s highest-scoring incorrect class, the non-constant family contributes a positive margin on every example in both branches, including every example the masked model answers wrongly, and on those examples its margin is higher, +6.08 against +5.30. The failures coincide with stronger opposition from the remainder: the remainder contributes negatively on more than nine tenths of examples in both branches, and in the control an opposition of -6.43 outvotes a task-aligned term of +5.72. Rescaling the task-aligned component by α , with nothing retrained,

raises the masked model monotonically on the sampled grid from 45.85% to at least 95% at $\alpha = 2.55$ and 99.91% at $\alpha = 6$: the intervention changes the balance of the two logit contributions without retraining. The rescaled component is $P_{\text{add}}h - \bar{h}$, with $\bar{h}$ the global mean. Recovery therefore depends on the task structure supplied by the projection. Family power share is a descriptive measure of this change.

Family power during grokking. Along an AdamW trajectory sampled every thousand steps, the addition family reaches 95.14% full-grid filtered accuracy at step 3,000, when the model’s held-out test accuracy is 0.30% and the family holds 1.05% of non-constant power. Ablating the family leaves near-chance accuracy at every sampled checkpoint. Its share rises to 70.62% at the test-accuracy jump, alongside greater concentration within the family. The memorizer control shows why this filtered accuracy alone cannot establish algorithm learning at step 3,000. Increasing power share describes the trajectory and is compatible with the cleanup account of Nanda et al. (2023). The margin decomposition and rescaling intervention establish amplitude competition between projected logit components at the masked checkpoint.

Spectral stability and functional change. Exact nonzero Fourier support is invariant under an invertible residual-basis map; power and its rankings are invariant under orthogonal maps. General invertible maps can change dominant-frequency membership. A jointly compensated basis change preserves predictions. At the collapse in Figure 3, dominant-frequency Jaccard is 1.0000 and power cosine is 0.9899, while the filtered pathway’s mean margin changes from +18.14 to −46.85. Over this interval the unembedding moves by 1.285 of its norm and the addition-filtered residual by 0.392; the old head on new features scores 26.31%, against 5.74% for the reverse. Adjacent swaps provide finer localization (Figure 1a). Accuracy returns to 99.90% within 300 steps with a changed power allocation. In the depth-4 pair, patching the frozen branch’s addition component of the final MLP write into the control gives 99.99% with the frozen head and 5.05% with the control head. These results show readout sensitivity; exact basis equivalence requires the separate tests in Appendix J.2.